



Tansley review

Eco-evolutionary optimality as a means to improve vegetation and land-surface models

Author for correspondence:
Sandy P. Harrison
Email: s.p.harrison@reading.ac.uk

Received: 17 January 2021
Accepted: 25 May 2021

Sandy P. Harrison^{1,2} , **Wolfgang Cramer**³ , **Oskar Franklin**^{4,5} ,
Iain Colin Prentice^{2,6,7} , **Han Wang**² , **Åke Brännström**^{4,8} ,
Hugo de Boer⁹ , **Ulf Dieckmann**^{4,10} , **Jaideep Joshi**⁴ ,
Trevor F. Keenan^{11,12} , **Aliénor Lavergne**¹³ , **Stefano Manzoni**¹⁴ ,
Giulia Mengoli⁶ , **Catherine Morfopoulos**⁶ , **Josep Peñuelas**^{15,16} ,
Stephan Pietsch^{4,17} , **Karin T. Rebel**⁹ , **Youngryel Ryu**¹⁸ ,
Nicholas G. Smith¹⁹ , **Benjamin D. Stocker**^{20,21} and **Ian J. Wright**⁷

¹Department of Geography and Environmental Science, University of Reading, Reading, RG6 6AB, UK; ²Ministry of Education Key Laboratory for Earth System Modeling, Department of Earth System Science, Tsinghua University, Beijing 100084, China; ³Institut Méditerranéen de Biodiversité et d'Ecologie Marine et Continentale, Aix Marseille Université, CNRS, IRD, Avignon Université, Technopôle Arbois-Méditerranée, Aix-en-Provence Cedex 04, F-13545, France; ⁴International Institute for Applied Systems Analysis, Schlossplatz 1 Laxenburg 2361, Austria; ⁵Department of Forest Ecology and Management, Swedish University of Agricultural Sciences, Umeå 90183, Sweden; ⁶Department of Life Sciences, Imperial College London, Silwood Park Campus, Buckhurst Road, Ascot, SL5 7PY, UK; ⁷Department of Biological Sciences, Macquarie University, North Ryde, NSW 2109, Australia; ⁸Department of Mathematics and Mathematical Statistics, Umeå University, Umeå 901 87, Sweden; ⁹Copernicus Institute of Sustainable Development, Environmental Sciences, Faculty of Geosciences, Utrecht University, Vening Meinesz Building, Princetonlaan 8a, Utrecht 3584 CB, the Netherlands; ¹⁰Department of Evolutionary Studies of Biosystems, The Graduate University for Advanced Studies (Sokendai), Hayama, Kanagawa 240-0193, Japan; ¹¹Climate and Ecosystem Sciences Division, Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA; ¹²Department of Environmental Science, Policy and Management, University of California Berkeley, Berkeley, CA 94720, USA; ¹³Department of Physics, Imperial College London, South Kensington Campus, London, SW7 2AZ, UK; ¹⁴Department of Physical Geography and Bolin Centre for Climate Research, Stockholm University, SE-106 91, Stockholm, Sweden; ¹⁵CSIC, Global Ecology, CREAF-CSIC-UAB, Bellaterra, Barcelona, Catalonia 08193, Spain; ¹⁶CREAF, Cerdanyola del Valles Barcelona, Catalonia 08193, Spain; ¹⁷BOKU - University of Life Sciences and Natural Resources, Gregor-Medel-Strasse 33, Vienna 1180, Austria; ¹⁸Department of Landscape Architecture and Rural Systems Engineering, Seoul National University, Seoul 08826, South Korea; ¹⁹Department of Biological Sciences, Texas Tech University, 2901 Main Street Lubbock, TX 79409, USA; ²⁰Department of Environmental System Science, ETH, Universitätstrasse 2, Zürich CH-8092, Switzerland; ²¹Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Zürcherstrasse 111, Birmensdorf 8903, Switzerland

Contents

Summary	2126	V. Outstanding issues	2135
I. Introduction	2126	VI. Concluding remarks	2136
II. Model development: problems and solutions	2127	Acknowledgements	2137
III. Leaf-level and canopy-level optimality	2128	References	2137
IV. Beyond the leaf level	2134		

New Phytologist (2021) **231**: 2125–2141
doi: 10.1111/nph.17558

Key words: acclimation, eco-evolutionary optimality, global vegetation model, land-surface model, leaf economics spectrum, plant functional ecology, stomatal behaviour, water and carbon trade-offs.

Summary

Global vegetation and land-surface models embody interdisciplinary scientific understanding of the behaviour of plants and ecosystems, and are indispensable to project the impacts of environmental change on vegetation and the interactions between vegetation and climate. However, systematic errors and persistently large differences among carbon and water cycle projections by different models highlight the limitations of current process formulations. In this review, focusing on core plant functions in the terrestrial carbon and water cycles, we show how unifying hypotheses derived from eco-evolutionary optimality (EEO) principles can provide novel, parameter-sparse representations of plant and vegetation processes. We present case studies that demonstrate how EEO generates parsimonious representations of core, leaf-level processes that are individually testable and supported by evidence. EEO approaches to photosynthesis and primary production, dark respiration and stomatal behaviour are ripe for implementation in global models. EEO approaches to other important traits, including the leaf economics spectrum and applications of EEO at the community level are active research areas. Independently tested modules emerging from EEO studies could profitably be integrated into modelling frameworks that account for the multiple time scales on which plants and plant communities adjust to environmental change.

1. Introduction

The ability of land ecosystems to deliver societal benefits – including the regulation of climate, the carbon cycle and water and air quality, and the provisioning of goods including food and fibre – is at risk because of current rates of global environmental change (Ostberg *et al.*, 2018). Assessing and mitigating this risk requires the reliable characterisation of vegetation processes, including plant demography, growth and competition as well as physical land–atmosphere interactions, at multiple spatial and temporal scales. Highly developed, process-based computational models now exist that operate across scales, simulating photosynthesis and stomatal regulation, carbon allocation, competition for light, water and nutrients, community assembly, disturbance regimes, interactions of vegetation with climate and atmospheric composition, and yields of essential products including crops. The two main (overlapping) categories of current models are dynamic global vegetation models (DGVMs) and land-surface models (LSMs). LSMs are designed for embedding in climate models and represent ‘fast’ land–atmosphere exchanges explicitly, typically with half-hourly time-steps. Some LSMs treat vegetation composition and structure as static; others simulate vegetation dynamics as well and are therefore also DGVMs. Conversely, some ‘offline’ (i.e. not coupled to a climate model) DGVMs represent fast land–atmosphere exchanges implicitly, using daily time-steps. LSMs – with or without dynamic vegetation – provide the means for Earth system models (ESMs) to represent the land–atmosphere interface, including impacts of atmospheric CO₂ and climate change on vegetation and feedbacks from vegetation changes on CO₂ and climate.

Process-based global vegetation models, including DGVMs and LSMs without dynamic vegetation, are based on explicit formulations of a set of processes rather than on observed relationships (by contrast, for example, with forest yield tables) and they use generic plant types (by contrast, for example, with most crop models). Process-based vegetation models have been extensively applied by

the climate, integrated assessment and impacts modelling communities to assess the nature and impacts of projected climate change, including the role of biophysical and biogeochemical feedbacks. An ensemble of global vegetation models is used in the Global Carbon Project’s annual update on the state of the carbon cycle (Friedlingstein *et al.*, 2020). Model ensembles are widely used to assess the role of vegetation in land–atmosphere interactions, such as diagnosing the causes of fluctuations in the atmospheric CO₂ growth rate (Keenan *et al.*, 2016). Vegetation models have also been used to quantify the magnitude of the positive climate–carbon cycle feedback and the negative CO₂ fertilization feedback to climate (e.g. Cox *et al.*, 2013), to investigate the impact of recent climate change on the hydrological cycle (e.g. Ukkola & Prentice, 2013), and to project the impact of future climate change on crop production (Inter-Sectoral Impact Model Intercomparison Project, ISIMIP2b: <https://www.isimip.org/protocol/#isimip2b/>; Ostberg *et al.*, 2018).

Modelling vegetation as a fully embedded component of the climate system presents major scientific and computational challenges (Fisher & Koven, 2020). The many successful applications of vegetation models have drawn attention away from several known systematic failures, which have emerged especially when models have tried to reproduce large-scale phenomena encoded in atmospheric measurements. For example, both ESMs and offline DGVMs failed to reproduce the full magnitude of the amplification of the high-latitude seasonal cycle of atmospheric CO₂ over the past half-century (Graven *et al.*, 2013; Thomas *et al.*, 2016). DGVMs also failed to reproduce the observed relationship between the ¹³C/¹²C ratio of atmospheric CO₂ and global land–atmosphere carbon exchange (Peters *et al.*, 2018). There are persistent disagreements between models, even about the sign of the effect of global warming on primary production (Ciais *et al.*, 2013). There are large uncertainties in the modelled response of vegetation to precipitation changes (Huntzinger *et al.*, 2017) and little agreement in the simulated response to CO₂ and the role of nutrient availability in modulating this response (Wieder *et al.*,

2015). Large differences in the modelled behaviour of global vegetation, which have persisted for more than two decades (VEMAP, 1995; Friedlingstein *et al.*, 2006), were identified as a serious concern in the Intergovernmental Panel on Climate Change (IPCC) Fifth Assessment Report (Ciais *et al.*, 2013). Developments since then have not alleviated this concern (Arora *et al.*, 2020).

These problems suggest a need to re-assess the assumptions and processes included in current vegetation models, and the modalities by which they are developed. The explosion over the last 20 years in the amount and variety of data available – including plant trait databases, field campaigns, flux measurements, ecosystem experiments and satellite remote sensing – should greatly facilitate the process of developing better models. Indeed, meta-analyses of different types of observation have provided insights into universal patterns which can be used for testing general patterns in simulated ecosystem responses to various drivers (Wieder *et al.*, 2019). However, finding ways to test alternative hypotheses using observations will mean moving beyond meta-analysis. Large-scale field experiments provide under-utilised opportunities for model evaluation (Medlyn *et al.*, 2015), while controlled-environment experiments are irreplaceable for testing general hypotheses about plant function. Controlled-environment experiments could, for example, help to resolve current disagreements about the impact of changes in CO₂ or nutrients on photosynthetic traits.

Progress, however, also requires coherent, well motivated hypotheses to test. Franklin *et al.* (2020) highlighted recent developments that hold promise for improving vegetation models by generating such hypotheses. The concept of eco-evolutionary optimality (EEO) is one of these developments. EEO invokes the power of natural selection to eliminate uncompetitive trait combinations, and thereby shape predictable, general patterns in vegetation structure and composition. The term ‘eco-evolutionary’ expresses the fact that organisms adjust to their environment on both shorter (eco-physiological, days to months) timescales and longer (demographic and evolutionary) timescales.

Community-mean trait values are to some extent an outcome of acclimation by individual plants, but also of adaptation: environmental filtering ensures that genotypes and species are present only in environments that fall within their acclimation capacity. The extent to which the observed variability in plant functional traits is due to phenotypic plasticity (individual acclimation) or to nonplastic genotypic differentiation and species replacement (Meng *et al.*, 2015; Dong *et al.*, 2017, 2020; Yang *et al.*, 2018) is essential to understand how community function and composition react to rapid environmental changes. Plastic traits, such as photosynthetic capacity (Togashi *et al.*, 2018b) and the temperature optimum of photosynthesis (Kumarathunge *et al.*, 2019; Vico *et al.*, 2019), acclimate quickly (days to weeks) within individual leaves; while other leaf traits, such as leaf mass per area (LMA), show only partial within-species adjustment to changes along environmental gradients (Dong *et al.*, 2017, 2020). Hydraulic traits, particularly leaf hydraulics linked to LMA and wood properties in general, also show limited plasticity. Adaptive changes in the mean abundance of nonplastic traits can only occur through the slower processes of community dynamics, which depend on demography and competition among species.

Eco-evolutionary optimality hypotheses are based on identifying trade-offs that organisms are required to make, for example in land plants between CO₂ uptake and water loss, and expressing these mathematically. At the core of modelling EEO are therefore the mechanistic links between plant functional traits, their implications for resource demand and acquisition and biogeochemical cycling, and their effect on the plant's competitiveness. Process-based vegetation models are suited to resolve these links and thus provide a useful framework for investigating how EEO shapes global vegetation function and climate-land feedbacks in the Earth system. EEO hypotheses have shown a notable ability to predict observed patterns, providing parsimonious explanations of observed traits at the leaf (e.g. Smith *et al.*, 2019; H. Wang *et al.*, 2020), plant (e.g. Farrior *et al.*, 2013; Lavergne *et al.*, 2020b) and vegetation (e.g. Franklin *et al.*, 2014; Baskaran *et al.*, 2017) levels. However, there is no recipe to generate a ‘correct’ EEO criterion. EEO formulations must be assessed against data, like hypotheses in all fields of science.

Many modelling groups are exploring the use of EEO hypotheses to improve the representation of specific processes in vegetation models (e.g. Bonan *et al.*, 2014; De Kauwe *et al.*, 2015; Lin *et al.*, 2015; Ali *et al.*, 2016; Xia *et al.*, 2017; Lawrence *et al.*, 2019). In this review, we aim to raise awareness of the broader potential for a hypothesis-testing approach based on EEO to underpin a more far-reaching improvement in the robustness and reliability (*sensu* Prentice *et al.*, 2015) of vegetation models. Section II provides a perspective on the shortcomings of current models and model development practice, informed by our collective experience. Section III introduces case studies that exemplify how EEO can improve (and often, simplify) formulations of core processes at the leaf level that are required by both LSMs and DGVMs. Section IV considers the scope for applying EEO at the whole-plant and plant community levels. Section V deals with limitations of the EEO concept. Section VI briefly considers the outlook for next-generation vegetation models incorporating EEO principles.

II. Model development: problems and solutions

The origins and historical development of global vegetation models have been reviewed by Prentice *et al.* (2007), Quillet *et al.* (2010), Prentice & Cowling (2013) and Fisher *et al.* (2014); this material will not be revisited here. Current models have much in common. Processes are differentiated by operational time-steps: canopy-atmosphere energy exchanges and photosynthesis are modelled in LSMs typically at half-hourly time-steps; phenology, carbon allocation and growth at time-steps of days to months; and vegetation dynamics and disturbance in DGVMs at time-steps of months to years. Most models represent plant adaptations to environmental conditions by specifying a limited set of PFTs, each characterised by a distinct set of attributes. This is problematic because (a) for most quantitative plant traits, variation is greater within than between PFTs (Kattge *et al.*, 2020), and (b) a substantial fraction of the observed variation in community-mean trait values along environmental gradients is linked to acclimation and adaptation within species and PFTs (Siefert *et al.*, 2015). Furthermore, incorporation of new processes within this framework necessarily implies a proliferation of new PFTs and the

necessity to derive estimates of parameter values for each new PFT. An alternative approach that has been gaining traction over the past decade has been towards the simulation of quantitative traits that vary dynamically, mimicking acclimation and/or adaptation processes (van Bodegom *et al.*, 2012) and more realistically portraying ecosystem carbon uptake (Verheijen *et al.*, 2015) and the dynamic response of terrestrial ecosystems to climate change (Reichstein *et al.*, 2014; Sakschewski *et al.*, 2015).

Several problems (see Box 1) however are slowing progress. Some of these may have arisen because of the general institutional separation of model development from empirical science, and the accretional nature of much of this development. Some are practical issues about the way model codes are written, updated and tested. All could be mitigated by adopting different practices (see Box 1). Several initiatives have promoted systematic data-model comparison ('benchmarking') of land models as a partial remedy for these problems (<https://www.ilamb.org/>; Collier *et al.*, 2018). Some proposed benchmark data sets – for example for upscaled gross primary production (Tramontana *et al.*, 2016) and plant litter properties (Pettinari & Chuvieco, 2016) – however are themselves modelled outputs, which limits their usefulness. Wieder *et al.* (2019) drew attention to the limitations of benchmarking, which (we suggest) should be considered as a necessary but by no means sufficient part of model evaluation. Wieder *et al.* (2019) described the recent history of a leading LSM, the Community Land Model (CLM), in particular its evolution from version 4 (a pioneering attempt to include the coupling of ecosystem carbon and nitrogen cycles, which however greatly underestimated global land carbon uptake and its sensitivity to climate) to the better-performing version 5. They made a strong case for the need to use observational and experimental information as an integral part of model development and improvement. Although experimental and observational studies in this field are often justified by the need to improve vegetation models, the pathway by which this new knowledge is transferred is indirect. In principle, experiments could be performed precisely to clarify how individual processes should be represented in models. This is still far from being standard practice.

A further consequence of the current model development paradigm is the complexity trap (Franklin *et al.*, 2020). Many recently published model 'improvements' are achieved by adding complexity (Fisher & Koven, 2020), but it is generally understood that this does not equate to increased realism, particularly as the incorporation of new processes often increases further the number of poorly known parameters that need to be specified. Moreover, developing models by accretion has inevitably led to a decline in transparency (Prentice *et al.*, 2015). In other areas of environmental modelling, including climate modelling, there has been a growing realisation that re-examination of basic processes, reduction of complexity and increased transparency are all necessary for progress (Held, 2005; Gramelsberger *et al.*, 2020). Vegetation modelling is no exception.

Different Earth subsystems, however, pose specific challenges. The key challenge for global-scale modelling of biological systems is to identify principles applicable across diverse and phylogenetically distinct assemblages (Franklin *et al.*, 2020). EEO could have a key role to play because it can generate coherent, testable hypotheses

Box 1 How surface land models are developed: some problems and proposed solutions.

Models do not encapsulate a coherent body of theory. Current models represent the accumulation of work by successive cohorts of scientists in a specific institution or collaboration. Generally, no one has a complete overview of what they contain, or why particular decisions were made in the past (Fisher & Koven, 2020). This approach is normal for operational (e.g. national weather or air-quality forecast) models, but it inhibits efforts to trace why a model behaves as it does. We suggest there is a need for a *re-design*, in which the core processes are examined one-by-one and specific hypotheses about these processes tested against relevant data. Our proposed theory-driven approach more closely resembles how quantitative models are used in most other fields of science.

Lack of clarity about hypotheses. Many aspects of plant and ecosystem function are subject to alternative interpretations, and it seems likely that some differences among the results of models originate in different hypotheses adopted for one or another process. These hypotheses are seldom explicitly stated, however. Although model codes and documentation are now commonly made public (a positive development), the models' complex history can make it hard to discern their underlying logic. There is a strong argument for greater clarity, and above all, a *re-examination of the evidence* underlying the representation of each process.

Unclear testing protocols. New process representations are often assessed by changing one component within an existing, complex model and examining the effect on model outputs. Results obtained in this way are seldom clear cut, however, because they are potentially vitiated by errors elsewhere in the model. Instead, *model components should be tested independently of others*.

Core process representations tend to be conserved. In many vegetation models, representations of the core processes of coupled energy, carbon and water exchange have remained unchanged since their original formulation. There is, in principle, no reason why new representations of core processes should not replace existing ones. However, it is noteworthy that this has not generally happened. The 'legacy' formulations were likely provisional, and might now be obsolete, yet they are seldom isolated and tested, while model 'improvement' more often consists of adding new processes (Fisher *et al.*, 2014). We propose that model development should be *re-focused on the critical analysis and evaluation of core process representations*, and new processes added only if evidence unambiguously shows that they are required.

Neglect of available observations. Model parameter estimation tends to be *ad hoc* and is frequently based on single values for 'model' species that are long outdated. Although there have been efforts to use available trait databases for defining PFT-specific parameterisations (e.g. Harper *et al.*, 2016), models are still relatively uninformed by the wealth of currently available observations. This situation could be remedied by *closer integration of data analysis and experimental research* into model development.

about plant and vegetation function that transcend differences among biomes and floras.

III. Leaf-level and canopy-level optimality

In this section, we summarise some case studies that demonstrate where EEO approaches have provided parsimonious representations of core, leaf-level processes that are individually testable and

supported by evidence. The case studies are presented roughly in descending order of readiness from photosynthesis and primary production, where a proof-of-concept for implementation in a LSM framework exists, to the leaf economics spectrum, which requires a novel approach to account for how phylogeny and environment co-determine plant traits.

1. Photosynthesis and primary production

Nearly all LSMs and DGVMs simulate photosynthesis using the Farquhar, von Caemmerer and Berry (FvCB) model (Farquhar *et al.*, 1980) or the modification proposed by Collatz *et al.* (1991). Implementing the FvCB model in its original form requires three parameters that are known to vary: the maximum carboxylation rate (V_{cmax}), which determines the enzymatic capacity for carbon fixation; the maximum electron-transport rate (J_{max}), which determines the capacity to generate the required reducing power; and the ratio of leaf-internal to ambient CO_2 ($c_i : c_a$, here denoted as χ), which relates the assimilation rate to stomatal conductance. An improved understanding of how V_{cmax} and J_{max} vary with environmental conditions should provide a more rigorous basis for modelling photosynthesis and primary production (Rogers *et al.*, 2017). Light-use efficiency (LUE) models – widely used in remote-sensing applications – simulate primary production using empirical response functions that modify the assumed proportional relationship between gross primary production (GPP) and light absorbed by the canopy (I_{abs}). This approach can limit the number of parameters to be estimated but severs the link to processes.

The model described in Box 2 predicts a number of related physiological characteristics correctly, including the global pattern of V_{cmax} in relation to light, temperature and vapour pressure deficit (VPD) (Smith *et al.*, 2019), seasonal variations of V_{cmax} across diverse ecosystems (Jiang *et al.*, 2020), elevational trends in photosynthetic traits and primary production (Peng *et al.*, 2020), and the response of V_{cmax} to atmospheric CO_2 (Smith & Keenan, 2020). Specifically, the model predicts a decline in V_{cmax} with increasing ambient CO_2 (Wang *et al.*, 2017), and a steeper increase with decreasing ambient CO_2 . Both have been verified experimentally (Fig. 1).

Neglecting the adaptive adjustment of V_{cmax} to growth conditions could result in simulated PFTs becoming (unrealistically) maladapted to environmental changes, and (if set too low) unrealistically steep responses of photosynthesis to temperature and ambient CO_2 . The model for GPP outlined in Box 2 provides an example of how EEO hypothesis formulation and testing can lead not only to a more realistic representation of a key process, in the sense of being well supported by observational and experimental data, but also to a less complex one. Compared with conventional models, the number of parameters required as input has been dramatically reduced, by two mechanisms. First, the adaptive adjustment of key quantities (V_{cmax} , J_{max} , and ξ , which determines the response of χ to VPD) eliminates the need for these to be prescribed. Second, it removes the need to provide lists of parameter values for PFTs.

This model is not suitable for immediate incorporation into a LSM because it works on multiday time-steps (i.e. at the time scale

of leaf- and canopy-level acclimation). However, the fast responses of photosynthesis and stomatal conductance to environmental variations are already well characterised (Farquhar *et al.*, 1980; Medlyn *et al.*, 2011). All that is needed, then, is to replace fixed values of V_{cmax} and J_{max} at a standard temperature (e.g. $V_{\text{cmax}25}$, $J_{\text{max}25}$) and ξ with slowly time-varying values that follow the EEO criteria. This is straightforward in principle and allows the same model to reproduce measured daily cycles of GPP in different biomes with unchanged parameter values (Fig. 2; Mengoli *et al.*, 2021). Moreover, whereas adding a new process (acclimation of photosynthetic parameters) might be expected to increase model complexity, this example shows the opposite: treating acclimation as an EEO process allows GPP to be represented more parsimoniously than would otherwise have been possible.

2. Dark respiration

Leaf mitochondrial respiration supports many metabolic processes, including the protein turnover required to maintain photosynthetic capacity. Leaf dark respiration (R_{dark}) is a widely measured quantity. Its instantaneous temperature response is commonly represented by the Arrhenius equation (Atkin *et al.*, 2017). Many models assume that R_{dark} at 25°C ($R_{\text{dark}25}$) is proportional to $V_{\text{cmax}25}$, treat this as a constant per PFT and model the temperature dependence of R_{dark} and V_{cmax} with separate Arrhenius equations. However, there is considerable spatial and temporal variability in R_{dark} within PFTs as a function of environmental conditions (Atkin *et al.*, 2015; Smith & Dukes, 2018). In a global analysis, H. Wang *et al.* (2020) showed that values of R_{dark} at current growth temperature are optimised according to the need to ensure that its metabolic functions are coordinated with V_{cmax} . This EEO hypothesis predicts that acclimated values of both R_{dark} and V_{cmax} increase with growth temperature, but less steeply than their instantaneous responses to temperature. These predictions are well supported by data; the conventional modelling approach is not (Fig. 3).

Neglecting the acclimation of leaf-level respiration is likely to be a major source of uncertainty in model predictions, with serious consequences for the estimation of land carbon uptake especially in warmer climates (Huntingford *et al.*, 2017). As with photosynthesis, there is no obstacle in principle to including leaf-level respiratory acclimation in DGVM or LSM frameworks. To do so requires only that $R_{\text{dark}25}$ varies along with (slowly varying) $V_{\text{cmax}25}$ following H. Wang *et al.*'s (2020) EEO hypothesis, while the fast environmental responses of R_{dark} and V_{cmax} are represented as in current models (or better, for R_{dark} , via the universal temperature response reported by Heskell *et al.*, 2016). Such a scheme has not been implemented yet, as far as we are aware, in any vegetation model.

3. Stomatal behaviour and transpiration

Plants regulate water and energy exchanges with the atmosphere by adjustment of stomatal conductance (g_s). Most current models represent g_s based on the fast, experimentally observed response to VPD (Damour *et al.*, 2010). More mechanistic models have been developed (e.g. Sperry *et al.*, 2017), but require new parameters (Drake *et al.*, 2017). EEO hypotheses, based on the trade-off

Box 2 Steps towards a parsimonious model for gross primary production.

We summarise here how EEO hypotheses were derived and tested and used to create a parsimonious model for GPP. For clarity, we describe a simplified model that assumes $J_{\max}$ is large. Wang *et al.* (2017) provide the full derivation.

(1) According to the FvCB model, photosynthesis proceeds at the lesser of two rates: A_C , determined by $V_{\max}$, and A_J , by light (with a maximum value determined by $J_{\max}$). The coordination hypothesis (Field & Mooney, 1986; Chen *et al.*, 1993; Maire *et al.*, 2012; Quebbeman & Ramirez, 2016; Wang *et al.*, 2017) states that these rates should converge. This is optimal in the sense that resources would be wasted if overcapacity were maintained in one process or the other. Therefore, the hypothesis predicts that $A_C \approx A_J$ on the time scale of physiological acclimation. This time scale, from empirical studies, is in the order of a few weeks. Assuming (as a simplification) that the response of A_J to light is linear over the relevant range, then from the FvCB model if $A_C = A_J$ then:

$$V_{\max}(c_i - \Gamma^*)/(c_i + K) = \varphi_0 I_{\text{abs}}(c_i - \Gamma^*)/(c_i + 2\Gamma^*) \quad \text{Eqn 1}$$

where c_i is the leaf-internal partial pressure of CO_2 , φ_0 is the intrinsic quantum efficiency of photosynthesis, and Γ^* and K are parameters (the photorespiratory compensation point and the effective Michaelis–Menten coefficient of RuBisCo, respectively) whose values, and dependencies on temperature and atmospheric pressure, are well established and relatively invariant across all C_3 plants (as also assumed by all global models that rely on the FvCB formulation of photosynthesis). Acclimation of $V_{\max}$ then ensures that photosynthesis follows the right-hand equation.

(2) We still need to know c_i . This depends only partly on the ambient CO_2 (c_a). The least-cost hypothesis is a conjecture by Wright *et al.* (2003) that the combined costs (per unit photosynthesis) of maintaining water transport and carbon uptake capacities are minimised, as carbon uptake requires water loss, and therefore transport. Prentice *et al.* (2014) reformulated this criterion based on the FvCB model, and proved that there is a value of the ratio $c_i : c_a$ (denoted χ) that minimises it. (See Dewar *et al.*, 2018 for extensions and alternatives.) The costs of water loss capacity were assumed proportional to transpiration, and the costs of carbon uptake capacity to $V_{\max}$. To minimise their sum, their derivatives must add up to zero:

$$a\partial(E/A)/\partial\chi + b\partial(V_{\max}/A)/\partial\chi = 0 \quad \text{Eqn 2}$$

where A is photosynthesis, E is transpiration, and a and b are (as yet) unknown quantities. $V_{\max}/A$ can be derived from the FvCB model. E/A can be derived from the diffusion equation (for CO_2 entering and water exiting the leaf, both controlled by stomatal conductance): $A = g_s c_a (1 - \chi)$ and $E = 1.6 g_s D$, therefore:

$$E/A = 1.6D/[c_a(1 - \chi)] \quad \text{Eqn 3}$$

where g_s is stomatal conductance (to CO_2 ; conductance to water is 1.6 times larger) and D is the leaf-to-air VPD. The solution to Eqn 2 is:

$$\chi = \Gamma^*/c_a + (1 - \Gamma^*/c_a)\xi/(\xi + \sqrt{D}) \quad \text{Eqn 4(a)}$$

with

$$\xi = \sqrt{b(K + \Gamma^*)/1.6a} \quad \text{Eqn 4(b)}$$

Eqn 4 allows us to derive c_i , which can be plugged into the right-hand side of Eqn 1.

(3) But what is the ratio b/a ? In principle a should decline with increasing temperature, as water becomes less viscous. So this ratio can be written β/η^* , where β is a parameter and η^* is the (known) viscosity of water relative to its value at 25°C. Wang *et al.* (2017) used global leaf stable carbon isotope data (a proxy for χ) to estimate a single, universal value for β by multiple regression. This is an approximation, of course, but Wang *et al.* (2017) could successfully predict the broad global patterns of χ ; how it varies with temperature, VPD and elevation; and how it varies among plant types, purely as a consequence of their growing in different environments.

(4) Eqn 1 predicts leaf-level photosynthesis. However, making the further assumption that the canopy behaves as a 'big leaf', Wang *et al.* (2017) and Stocker *et al.* (2020) showed that the same equation could be used to predict GPP, provided that the satellite-observed fractional absorbed photosynthetically active radiation (fAPAR) is used in the determination of I_{abs} . The model is an LUE model: that is GPP is proportional to I_{abs} . But now it is linked explicitly to the FvCB model. All of its parameters are independently known, or (like β) can be estimated from data independent of GPP. The full implementation (considering finite $J_{\max}$) requires one more parameter, but this too can be estimated from independent data (measurements of the ratio of $J_{\max}$ to $V_{\max}$).

between maximising carbon gain and minimising water loss, potentially offer parsimonious solutions. One approach (Medlyn *et al.*, 2011) is based on an approximate solution to the hypothesis of constant marginal water-use efficiency, originally proposed by Cowan & Farquhar (1977). This solution correctly predicts

stomatal responses to changing CO_2 and variability across environmental gradients (Medlyn *et al.*, 2011; Medlyn *et al.*, 2013; Lin *et al.*, 2015). It has been included in LSMs (e.g. De Kauwe *et al.*, 2015; Franks *et al.*, 2018; Oliver *et al.*, 2018) and shown to perform as well as the empirical relationships originally used in those

models. These approaches all require calibrating one 'free' parameter per PFT in the optimal stomatal conductance formulation. To achieve a parameter-free formulation, it is possible to re-

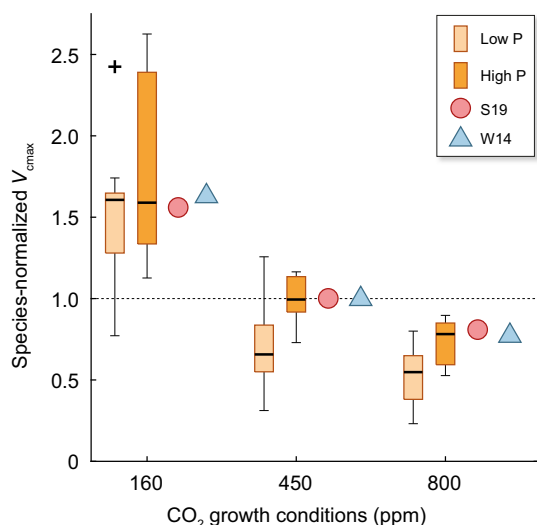


Fig. 1 Response of photosynthetic capacity (V_{cmax}) measured in *Holcus lanatus* (C_3 grass) and *Solanum dulcamara* (vine) to CO_2 , shown in growth-chamber experiments where ambient CO_2 and phosphorus (P) supply were manipulated (see Supporting Information Methods S1 and Table S1 for further details). Low-P and high-P treatments received fertiliser with nitrogen to phosphorus (N : P) ratios of 45 : 1 and 1 : 1, respectively. The box plots show the median (red line) and first and third quartile ranges, with whiskers showing the 5th and 95th percentiles, of the experimental data. Values outside these ranges are indicated with a red plus sign. The responses of V_{cmax} to subambient and elevated CO_2 are consistent with the predictions of the coordination hypothesis as modelled following the eco-evolutionary optimality formulations of Wang *et al.* (2014) (W14) and Smith *et al.* (2019) (S19). Experimental V_{cmax} was scaled relative to the high-P population average under ambient CO_2 growth conditions (450 ppm), whereas modelled V_{cmax} was scaled relative to the single 450 ppm prediction. The response to CO_2 is significant at the 99% confidence level, as is the response to P supply, but the interaction term is nonsignificant indicating that the response to CO_2 is the same regardless of P supply.

frame the Cowan and Farquhar (CF) hypothesis by accounting for soil moisture dynamics in the optimisation problem (Manzoni *et al.*, 2013), but this implies an assumption on how much soil water can be used by plants.

A different approach (Prentice *et al.*, 2014), used in the model of Wang *et al.* (2017) and summarised in Box 2, is based on the EEO hypothesis that leaves minimise the sum of the maintenance costs (per unit assimilation) of transpiration and carboxylation capacities. Carboxylation costs are envisaged as the respiration required for RuBisCo turnover, while transpiration costs are envisaged as the respiration required to maintain living sapwood. This hypothesis leads to a mathematical formulation of the fast response of stomata to VPD that is closely related to that of Medlyn *et al.* (2011), while also correctly predicting the environmental dependencies of the control parameter (ξ) on temperature (Lin *et al.*, 2015) and atmospheric pressure (Körner & Diemer, 1987). Each of these predicted partial dependencies of χ on environmental variables can be observed in stable carbon isotope ratio ($\delta^{13}C$) measurements on leaves (Wang *et al.*, 2017). Lavergne *et al.* (2020b; Fig. 4) showed they are also present in tree-ring $\delta^{13}C$ measurements. By providing time-series, Lavergne *et al.* (2020b) showed a (weak) dependency of χ on atmospheric CO_2 (Fig. 4) that is also predicted by this EEO hypothesis (Box 2). Apart from the well known VPD response, none of these dependencies is reflected in standard LSMs, except crudely, through the assignment of distinct parameter values to PFTs that occupy different climates.

Further alternative EEO approaches (e.g. Wolf *et al.*, 2016; Anderegg *et al.*, 2018; Eller *et al.*, 2018; Venturas *et al.*, 2018; Trugman *et al.*, 2019; Deans *et al.*, 2020; Eller *et al.*, 2020; Sabot *et al.*, 2020) incorporate hydraulic costs, based on the hypothesis that the short-term and long-term costs of transpiration at low soil water potential contribute to the total cost of maintaining the water transport pathway. The Wolf *et al.* (2016) formulation has been shown to be in good agreement with experimental evidence for changes in stomatal conductance in response to daily and seasonal changes of environmental conditions, and to perform much better

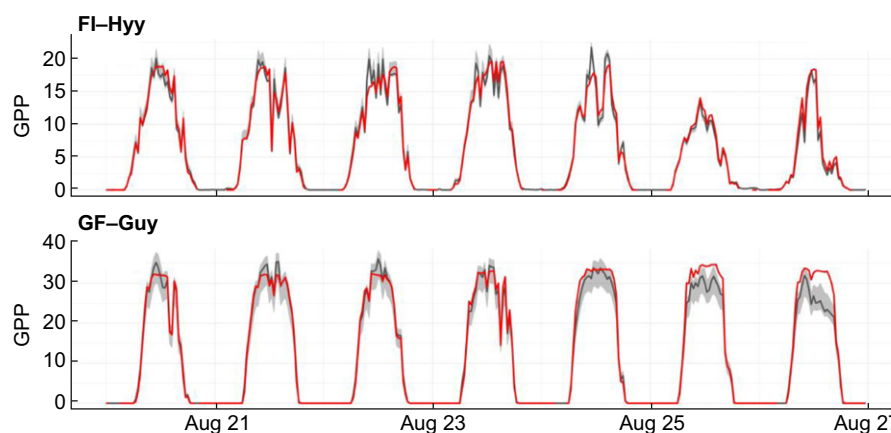


Fig. 2 Observed (by eddy covariance, GPPo, grey) and modelled (using the same parameter set, GPPp, red) half-hourly gross primary production (GPP) during 1 wk in August 2014 at sites in a tropical (GF-Guy) and a boreal (FI-Hyy) forest. Grey bands represent the uncertainty in GPP calculated using the daytime partitioning method in the FLUXNET2015 dataset (Pastorello *et al.*, 2020). Modelled and observed GPP are in units of $\mu\text{mol } CO_2 \text{ m}^{-2} \text{ s}^{-1}$. Figure redrawn based on analyses in Mengoli *et al.* (2021).

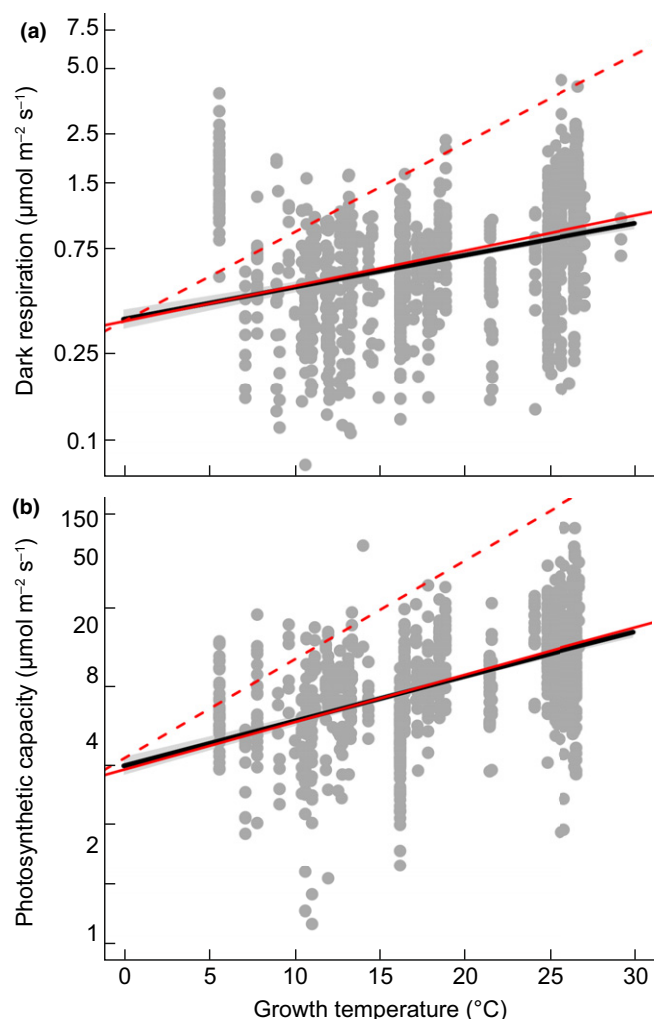


Fig. 3 Field-measured (black lines) (a) leaf dark respiration rates (R_{dark}) and (b) photosynthetic capacities (V_{cmax}) compared with their modelled responses to growing-season temperature (red solid lines) as predicted by the coordination hypothesis (H. Wang *et al.*, 2020). Both R_{dark} and V_{cmax} have been corrected (using the Arrhenius equation, with ΔH as provided by Bernacchi *et al.*, 2001) from the specific measurement temperature to the growing-season average temperature for the site. The coordination hypothesis predicts acclimated responses to temperature. The red dashed lines show the *instantaneous* responses to temperature, that is the relationship assumed by most models, based on the same Arrhenius equation. Data from the GlobResp (Atkin *et al.*, 2015) and LCE (Smith & Dukes, 2017) data sets. Figure redrawn from analyses presented in H. Wang *et al.* (2020).

than the CF model in predicting stomatal responses to dry soils (Anderegg *et al.*, 2018). Y. Wang *et al.* (2020) found that among 11 EEO-based stomatal models, the most skilled were those taking into account the cost of stress-induced hydraulic failure.

The development of EEO hypotheses for stomatal behaviour is a highly active research field. The hypothesis of Prentice *et al.* (2014) provided an equation with good predictive power for the responses of leaf and plant $\delta^{13}\text{C}$ to the growth environment, but the one ‘universal’ parameter it includes has been shown to be influenced by soil moisture (Lavergne *et al.*, 2020a) and soil pH (Wang *et al.*,

2017; Paillassa *et al.*, 2020). Moreover, the variation of χ on long climatic moisture gradients appears to be significantly steeper than predicted by that equation (Dong *et al.*, 2020). Allowing variation of ξ following the EEO criterion of Prentice *et al.* (2014) would allow stomatal acclimation to changes in growth temperature (Marchin *et al.*, 2016). This cannot happen in current models, because their responses to VPD are predetermined by PFT. However, further research is needed to determine how soil influences might best be included in models. Solutions are likely to include EEO approaches to explain the coordination of hydraulic and photosynthetic traits (Brodrribb, 2009; Deans *et al.*, 2020; Joshi *et al.*, 2020), and the influence of soil fertility factors on water and nutrient acquisition costs (Paillassa *et al.*, 2020).

4. Isoprene emission

Plant emissions of the volatile organic compound (VOC) isoprene protect photosynthetic function against damage due to reactive oxygen species (ROS), which are produced in leaves at high temperatures (Niinemets, 2010; Harrison *et al.*, 2013; Lantz *et al.*, 2019). As a result, tree species that produce isoprene are competitively favoured under hot and dry conditions (Taylor *et al.*, 2018). Modelling of plant VOC emissions is important in ESMs, because these reactive compounds have a significant impact on atmospheric chemistry. Many ESMs rely on a complex empirical model (Guenther *et al.*, 2006) to predict VOC emissions. More explicitly process-based models of VOC emission have been devised (e.g. Pacifico *et al.*, 2011) but still require several, poorly known parameters to be specified.

However, the responses of isoprene emission to light, temperature and CO_2 are consistent with a much simpler relationship: a linear dependency on the leaf’s ‘energetic status’, which is the difference between photosynthetic electron transport and the electron requirement to support the current rate of carbon fixation (Morfopoulos *et al.*, 2013). The coordination between these two rates is what enables plants to optimise LUE at low levels of irradiance and to dissipate excess energy as heat at high irradiance. The linear dependency on the energetic status reproduces the shapes of observed responses of isoprene emission to environment, including its nonlinear response to light (Fig. 5); its (otherwise enigmatic) decline with instantaneous increases in ambient CO_2 (Morfopoulos *et al.*, 2013, 2014); and its recovery over time at high CO_2 (Sun *et al.*, 2013). Without this recovery, isoprene emitters under high CO_2 would lose the thermoprotective benefits of isoprene emission, an unlikely outcome in evolutionary terms.

The leaf energetic status model has the potential to simplify the representation of VOC emission in ESMs, while increasing its predictive power for conditions outside those currently observed. However, a key unanswered question remains, regarding the ‘base rate’ of isoprene emission at the plant-community level. The current standard approach relies (unsatisfactorily) on a fixed geographic field of emission capacity. Explicit hypotheses to predict VOC emission capacity remain to be formulated and tested.

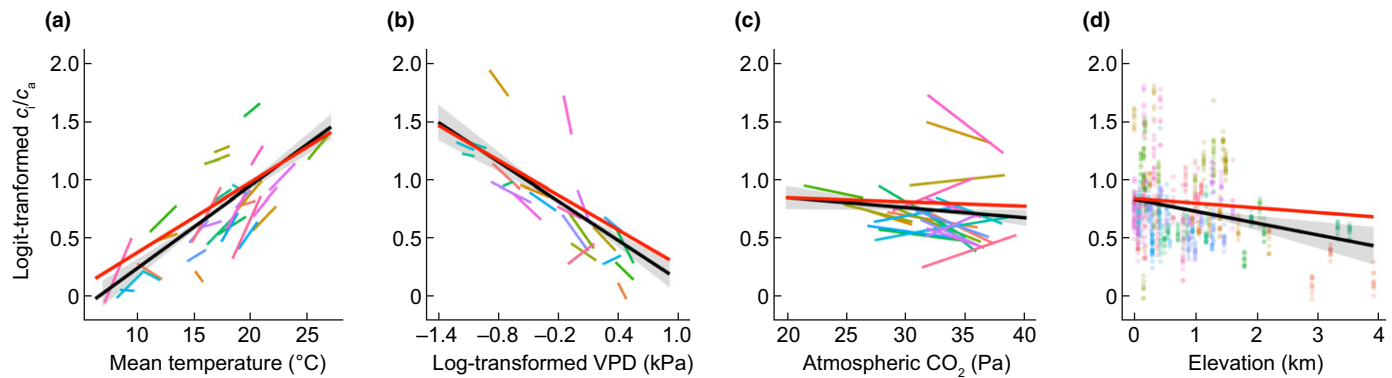


Fig. 4 Trends in the ratio of leaf-internal to ambient CO_2 reconstructed from stable carbon isotope ratios in tree rings for different sites (coloured lines) and for the whole dataset (black, with standard error shown by the grey shading) with respect to (a) mean growth temperature, (b) vapour pressure deficit (VPD), (c) atmospheric CO_2 and (d) elevation, compared with modelled trends (red line) for the whole dataset based on the least-cost hypothesis. Figure redrawn from data and model results described in Laverne *et al.* (2020b).

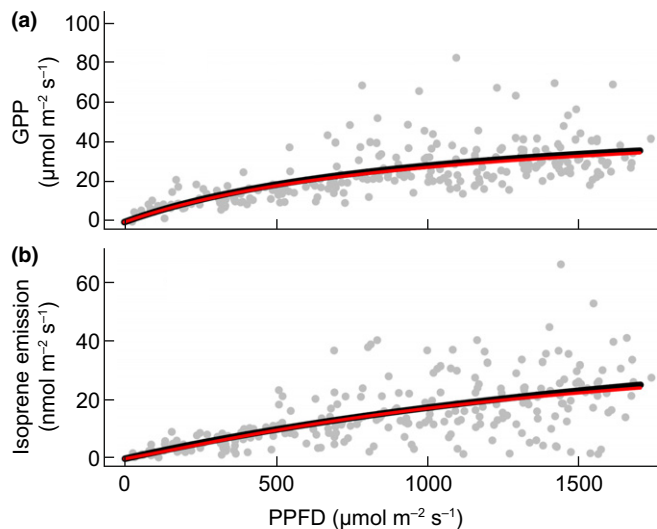


Fig. 5 Observed (a) photosynthesis and (b) isoprene emission at University of Michigan Biological Station forest in relation to incident photosynthetic photon flux density (PPFD), compared with modelled values (red lines) based on the FvCB model and the hypothesis that isoprene emission is related to leaf energetic status (Morfopoulos *et al.*, 2013). Data are shown for an air temperature range 24.5–25°C to limit the impact of temperature on the response to PPFD. The black lines represent relationships of GPP and isoprene emission rate to changes in PPFD estimated by Michaelis–Menten type functions fitted to the data. The figure is redrawn from Morfopoulos (2014) and the model is described in Morfopoulos *et al.* (2014).

5. Leaf economics

Leaf mass per unit area (LMA) determines how much leaf area can be produced for a given total carbon allocation to leaves (Cui *et al.*, 2019). The leaf economics spectrum (Wright *et al.*, 2004) relates LMA and leaf lifespan (LL) across vascular plant species. This relationship is not fixed, however, and varies with climate. Kikuzawa (1991) hypothesised that LL maximises leaf carbon gain over the lifetime of the leaf, accounting for (amortised) leaf construction costs. This EEO hypothesis has recently been combined with two others. Xu *et al.* (2017) provided empirical support for the hypothesis that the leaf ageing rate (a parameter in

Kikuzawa's model) is inversely proportional to LMA, and directly proportional to $V_{\text{cmax}25}$; while the coordination hypothesis, described above, predicts optimal values of $V_{\text{cmax}25}$. Combining these three EEO hypotheses leads to a theoretical prediction of the leaf economics spectrum, and how it varies across environments (Wang *et al.*, 2021). For winter-deciduous woody plants where LL is constrained by the length of the growing season, this theory leads to a prediction of LMA that is consistent with observations along an elevational and aridity gradient (Xu *et al.*, 2021). For evergreen plants it leads to a correct global prediction of the proportionality between LMA and LL, and how this is modified by growing-season length and light (Wang *et al.*, 2021; Fig. 6). A changing climate will inevitably alter the competitive balance among species with different LMA and LL, in ways that fixed LMA values per PFT cannot capture.

In addition to affecting LL, LMA mechanistically affects stomatal response (Buckley *et al.*, 2015). Increasing LMA reduces the conductivity of the outside-xylem water pathway due to increased path length, and therefore causes highly negative water potentials near the stomata. This in turn may necessitate a greater investment in leaf hydraulics in high-LMA species. Without such investment, these species would be uncompetitive due to reduced photosynthesis rates. Therefore, EEO suggests a testable linkage between physiological and hydraulic traits.

The EEO basis for the leaf economics spectrum has not been incorporated in any vegetation model, and its consequences for climate-change impacts are largely uncharted. LMA and LL, as structural/morphological traits, differ from the physiological traits discussed above in showing far less plasticity (or genotypic adaptation) along environmental gradients (Dong *et al.*, 2017, 2020). Therefore, their representation in models calls for a different treatment, as any change in community-mean LMA and LL will depend more on species replacement (a slower process) than on physiological adjustment. In addition, whereas the theory summarised above predicts environmentally conditioned changes in the *relationship* between LMA and LL, it does not predict anything about their mean values. Phylogenetic conservatism is helpful here. Starting from the observed global distributions of these traits, the model of Wang *et al.* (2021) calculates how these intersect with the

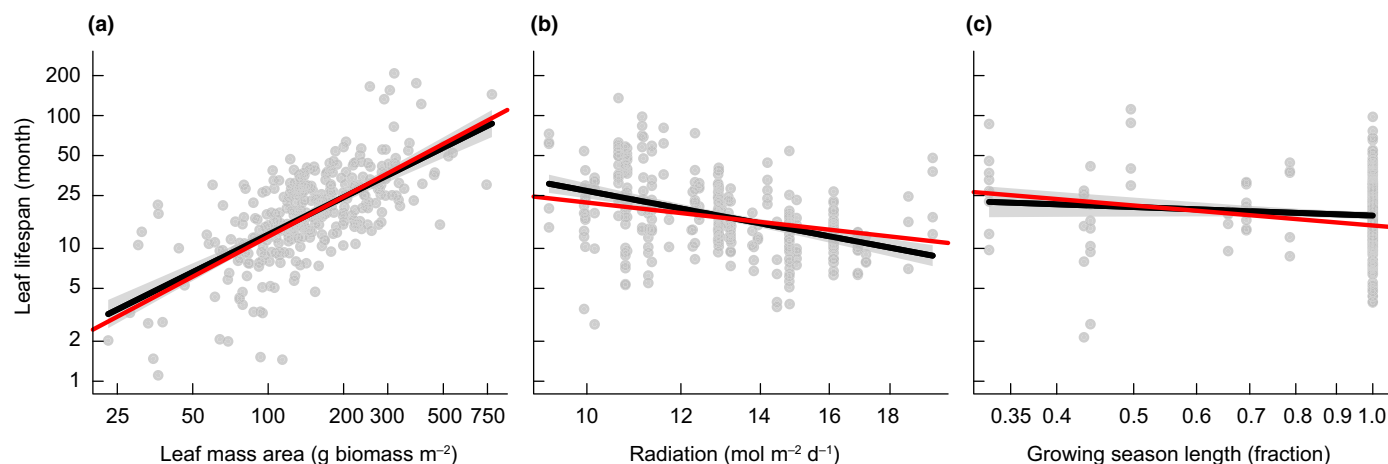


Fig. 6 Observed relationships (black lines) between LL and (a) leaf mass per area, (b) radiation and (c) growing-season length as a fraction of the year compared with relationships predicted (red lines) by the time-averaged maximisation of leaf carbon gain through leaf lifespan after accounting for construction costs. All values are shown on a log scale. Data from evergreen species in the GLOPNET trait database (Wright *et al.*, 2004). Figure redrawn from analyses presented in Wang *et al.* (2021).

predicted optimal LMA-LL relationship. This approach generates probability distributions around the predicted community-mean values as illustrated in Fig. 6.

IV. Beyond the leaf level

Most published applications of EEO concepts in vegetation modelling have focused on leaf-level physiological processes, facilitated by their phenotypic plasticity and the short timescale of some leaf-level responses to environmental conditions. The EEO framework however extends naturally to phenotypic plasticity at the whole-plant level, providing insightful approaches to modelling processes including phenological timing (Caldararu *et al.*, 2014; Manzoni *et al.*, 2015) and strategy, and carbon allocation to leaves, stems and roots (e.g. Valentine & Mäkelä, 2012; Kvakić *et al.*, 2020). We summarise some cases below. The EEO approach can be extended to nonplastic trait variation if selection on these traits is not strongly frequency dependent, that is if the fitness of plants does not depend strongly on the frequency of traits among their conspecifics (Metz *et al.*, 2008). EEO concepts are particularly powerful for describing trait coordination for example among different plant organs (Deans *et al.*, 2020): when multiple traits optimally adapt to the environment, environmental variation leads to covariation between them. Such emergent relationships may provide the basis for modelling how evolved traits vary with environmental conditions without simulating the underlying physiological mechanisms (or evolutionary dynamics) through which optimal coordination is achieved. We briefly consider now the potential application of the EEO framework to modelling whole-plant processes, competition and ecosystem dynamics.

1. Carbon allocation

Both field and experimental data show that allocation to roots increases when nutrient supply is limiting, for example on infertile soils or in cold climates (Poorter *et al.*, 2012; Reich *et al.*, 2014; Gill

& Finzi, 2016). This observation is consistent with the long-established EEO hypothesis that plants, requiring multiple resources, allocate effort optimally so that no one resource is limiting to growth (Rastetter & Shaver, 1992; Thomas & Williams, 2014; Rastetter & Kwiatkowski, 2020). A plant-level allocation model based on this assumption has been used to explain the contrasting effects of elevated CO₂ on tree growth and nitrogen uptake and their dependence on soil nitrogen availability (Franklin *et al.*, 2009; Fig. 7). An EEO approach to carbon allocation has been adopted in at least one LSM (Xia *et al.*, 2017).

2. Soil–plant interactions and mycorrhizae

Carbon exchanges between plants and their symbionts can also be described using EEO principles. The effects of nutrient limitation are predictable based on the carbon costs of nitrogen uptake via different symbionts (Terrer *et al.*, 2018), which may in turn depend predictably on soil nutrient availability (Franklin *et al.*, 2014). The fraction of GPP allocated to sustain symbionts therefore becomes an outcome, rather than being imposed as an additional parameter (Baskaran *et al.*, 2017). Modelling soil–plant interactions explicitly in terms of the carbon cost of nitrogen acquisition has a significant impact on modelled primary production (Brzostek *et al.*, 2014) and has been adopted in at least one LSM (Shi *et al.*, 2016). Dynamically linking plants and microbes in a terrestrial biosphere model has been shown to improve predicted carbon and nitrogen dynamics across a gradient of vegetation stands varying in the abundance of trees with distinct (arbuscular and ectomycorrhizal) types of mycorrhizal interaction (Sulman *et al.*, 2017).

3. Competition and coexistence

If the fitness of plants is strongly influenced by competition with other plants in a way that depends on the frequency of their traits, game-theoretic extensions of the EEO framework such as adaptive

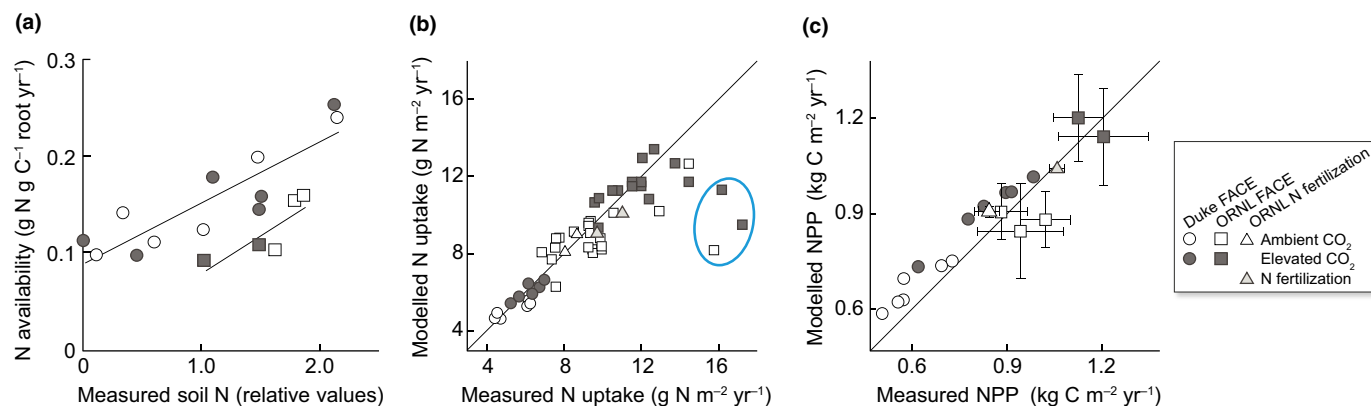


Fig. 7 Comparison of observed against modelled (a) soil N availability, (b) tree N uptake, and (c) net primary production (NPP) under ambient (open symbols) and elevated (dark grey symbols) CO₂, and with nitrogen fertilization (light grey symbols), at two free-air carbon dioxide enrichment experiments: Duke FACE (circles) and ORNL FACE (squares). The plant-level optimality model optimises canopy N, LAI, and fine-root production by maximising net growth as a proxy for fitness. The imposed treatment effects (independent variables) are light-saturated leaf-level photosynthetic N-use efficiency and soil N availability (maximal potential N uptake per root C). The r^2 is 0.90 for NPP, and 0.85 for N uptake (excluding circled outliers). Error bars in (c) show standard deviation across different years. Measured units of soil N in (a) are extractable nitrate (0.1 $\mu\text{g g}^{-1}$ soil) in Duke and net N mineralisation (20 $\mu\text{g g}^{-1}$ soil yr⁻¹) in ORNL. Figure redrawn from data and model results described in Franklin *et al.* (2009).

dynamics theory (Metz *et al.*, 1992; Dieckmann & Law, 1996) can be used to predict not only single optimal plant strategies but also co-existing and co-evolving strategies. Examples include the coexistence of different strategies for coping with water shortage in dry environments (Lindh *et al.*, 2014), and the complementarity of alternative life-history strategies generating within-site heterogeneity and corresponding variation in optimal strategies in resource-rich communities (Togashi *et al.*, 2018a). Falster *et al.* (2017) demonstrated the evolution of realistic patterns of stable coexistence of tree species in a height-structured competition model related to the demographic schemes used in DGVMs. Other game-theoretic approaches (e.g. Dybzinski *et al.*, 2015; Weng *et al.*, 2019) have shown that co-existing strategies can give rise to emergent trait coordination, which can be compared with empirical observations. This work offers the prospect of a wider field of application for EEO-based modelling to address the origins and maintenance of species diversity.

V. Outstanding issues

It should be abundantly clear from the discussions above that EEO, despite its utility, is not a 'magic bullet' that can instantly resolve problems in LSM and DGVM development. We suggest instead that EEO concepts should underpin a research programme in which explicit, quantitative hypotheses play a central role in data analysis and experimental design, while also providing parsimonious formulations for modelling. Practitioners of this integrative approach need to be aware of the limitations of EEO as well as its strengths. Some of the most important limitations are summarised below.

Natural selection acts on reproductive fitness, not on plant growth. However, reproductive fitness is very difficult to measure in plants in the field. EEO hypotheses can be formulated in terms of vegetative properties but it must be recognised that these are only indirectly linked to fitness. The underlying assumption is that

ineffective or uncompetitive trait combinations will confer low fitness and be selected against.

It follows from the above that *no EEO hypothesis is unique*. For every trade-off considered, there are alternative criteria all of which might appear to be compatible with EEO, but which make different predictions. (Some examples have been discussed above.) Only empirical tests can determine which, if any, of a series of alternative EEO hypotheses makes the most realistic predictions.

The limits to optimality are a priori unknown and can only be assessed empirically. Recent EEO approaches to photosynthetic optimality have made pragmatic choices in the interests of parsimony. For example, it has been assumed that certain photosynthetic traits can show unlimited variation, while others – such as the Michaelis constants and specificity of RuBisCo, the intrinsic quantum yield of electron transport, and their temperature dependencies – are treated as constants. These assumptions are supported by observations but only as an approximation; all these properties do in fact vary among plants (Ehleringer and Piercy, 1983; Dreyer *et al.*, 2001; Singsaas *et al.*, 2001; Galmés *et al.*, 2015; Galmés *et al.*, 2016), even if their variation is relatively modest.

Optimality is approached at different rates by different processes. In a realistically time-varying environment optimality is approached rather than achieved, because the optimum is a moving target and, indeed, competitive success does not necessarily require the optimum state to be achieved. We have distinguished the fast (minutes to hours) time scales of enzyme kinetics and stomatal responses to VPD from the slower time scales of leaf-level physiological acclimation (days to months), carbon allocation (months to years) and species replacement (years to decades). DGVMs respect these distinctions. However, the shift from a PFT to a trait basis for modelling necessitates ensuring that trait shifts dependent on species replacement take place on the appropriate time scale, which can be different for different traits. Again, empirical tests are important to determine whether the timescales of acclimation adopted for modelling purposes are realistic.

The problem of absent species. A harder issue related to time scales is how to represent dispersal and migration (in other words, species replacement when the best-adapted species are not locally present) in DGVMs. Although we do not review attempts to do so here, we note that an important test of existing approaches is whether the rates of migration they predict are consistent with observed rates of species replacement in response to rapid climate changes in the past (e.g. Harrison & Sanchez Goñi, 2010).

The importance of experiments. Comprehensive testing of EEO hypotheses cannot rely entirely on meta-analysis. Data from direct environmental manipulations are not hampered by correlations between environmental variables and can therefore be used to quantify the timescales of responses (Kumarathunge *et al.*, 2019). Controlled-environment glasshouse experiments have been used to determine the rates and mechanisms underlying acclimation of photosynthetic (e.g. Scafaro *et al.*, 2017), hydraulic (e.g. Locke *et al.*, 2013) and leaf-biochemical parameters (e.g. Dongsansuk *et al.*, 2013) to changes in the growth environment. Field experiments can scale individual to ecosystem-level responses, through direct manipulations (e.g. Hoeppepner & Dukes, 2012; Hovenden *et al.*, 2019), exclusion experiments (e.g. Inoue *et al.*, 2017; Tomasella *et al.*, 2018) or opportunistic sampling strategies (e.g. Lusk *et al.*, 2018). The increasing coordination of field experiments, including experiments to examine the impacts of manipulating nutrient (e.g. NutNet; Borer *et al.*, 2014) or water supplies (e.g. DROUGHT-NET; Knapp *et al.*, 2017), provides opportunities to evaluate the role of different plant strategies for coping with environmental stresses, and such networks provide key targets for model evaluation (e.g. Hilton *et al.*, 2019).

Fire and land use. Wildfire is a major cause of vegetation disturbance and many models now explicitly stimulate the two-way interaction of wildfire regimes with vegetation and climate. However, the performance of these models is relatively poor beyond the largest-scale geographic patterns (Forkel *et al.*, 2019; Hantson *et al.*, 2020). This raises the issue of whether there is scope for EEO concepts to inform research and ultimately improve fire-vegetation models. Plants have, for example, evolved specific adaptations to different frequencies and intensities of fire (Clarke *et al.*, 2013; Pausas *et al.*, 2016; Pausas, 2019). Exploring the trade-off between the maintenance cost of these traits and fire frequency could lead to more mechanistic representations of vegetation–fire interactions in models and the ability to project the consequences of environmental changes in fire-prone regions better.

The impact of changes in land use on the biophysical properties of the land surface and on biogeochemical cycling has led to considerable attention being given to developing data sets to impose land-use history on vegetation models (e.g. Pongratz *et al.*, 2008; Klein Goldewijk *et al.*, 2017) and scenarios for future changes in the land biosphere under direct human intervention (including agriculture, pastoralism and forestry). Several vegetation models now explicitly simulate agricultural PFTs to be able to assess the likely impact of future climate changes on production and the human resource base. Arable crops represent an extreme modification of the landscape yet, within the constraints that have been imposed by artificial selection, crop growth conforms to the same principles as all plants and can be modelled with the same EEO-

based tools (Qiao *et al.*, 2020). Further work to explore the EEO approaches to anthropogenic land use would be useful, both from the perspective of providing more realistic or more parsimonious crop models and to harmonise modelling approaches for simulating the land biosphere within ESMs.

VI. Concluding remarks

Vegetation models have shown their usefulness for projecting ecosystem productivity, vegetation patterns, terrestrial carbon uptake and other ecosystem services in a rapidly changing world. These projections now feed routinely into global assessments such as those being made by the IPCC, the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services and the Integrated Research on Disaster Risk project. Therefore, they contribute to the evidence base necessary to assess progress towards the United Nations Sustainable Development Goals (Heck *et al.*, 2018). However, more reliable models are required to enhance confidence in the plausibility of many of these projections. The rate of expected global environmental change increases the need for models to be able to deal with dynamic processes, including plant migration, adaptation, acclimation and land-use change. Global changes are occurring faster than many adaptive processes and are likely to result in novel environmental conditions; models must therefore be equipped to deal with nonequilibrium situations and novel conditions outside the range for which they were originally developed and tested. This can only be achieved by ensuring that they realistically account for acclimation and adaptation processes and do not entirely rely on statistically determined, historical patterns. However, increased realism is of little value if it is accompanied by over-parameterisation and ever-increasing parameter uncertainty. We have indicated how EEO theory can provide a means to alleviate these problems by substantially reducing the number of parameters required that must be specified. As models move away from PFTs to explicitly representing plant traits, EEO approaches will make it possible to reduce the dimensionality of the trait-space that needs to be considered. The application of EEO requires clear formulation of alternative hypotheses, which in turn creates a central role for observations and experiments to test and compare them.

There is as yet no comprehensive description of plant behaviour in terms of EEO – indeed, as some of the examples above have shown, the appropriate choice of optimality criteria is an active research topic in areas such as stomatal behaviour while in other areas, such as disturbance dynamics, EEO concepts are in their infancy. Moreover, there is no recipe for success. EEO hypotheses must be tested, and many will fail. EEO approaches are nonetheless providing robust, parsimonious and well supported representations of core processes that are represented in all vegetation models, and offer promise for the development of a new generation of models.

Acknowledgements

We gratefully acknowledge the contribution of participants at the workshop ‘Next generation vegetation modelling’, held at the International Institute for Applied Systems Analysis (IIASA) in March 2017. The idea for this review arose from the insights and

excitement engendered by these discussions. We thank IIASA, both for their financial support of the workshop, and for continued support thereafter. We particularly thank the previous Director General and CEO of IIASA, Pavel Kabat, for his support for the next-generation vegetation modelling initiative. SPH acknowledges support from the ERC-funded project GC2.0 (Global Change 2.0: Unlocking the past for a clearer future, grant number 694481). ICP, GM and CM acknowledge support from the ERC-funded project REALM (Re-inventing Ecosystem And Land-surface Models, grant number 787203). WC thanks the Labex OTMed (grant no. ANR-11-LABX-0061) funded by the French Government Investissements d'Avenir program of the French National Research Agency (ANR) through the A*MIDEX project (grant no. ANR-11-IDEX-0001-02). IJW acknowledges Australian Research Council funding (DP170103410). HW acknowledges support from the National Natural Science Foundation of China (no. 31971495) and the High End Foreign Expert awards at Tsinghua University to SPH and ICP (GDW20191100161). NGS acknowledges funding from Texas Tech University. JP acknowledges support from the ERC-funded project IMBALANCE-P (grant number 610028). AL was supported by a Marie Skłodowska-Curie Individual Fellowship (ECAW-ISO, grant number 838739). OF acknowledges funding provided by the Knut and Alice Wallenberg foundation. SM acknowledges funding from the Swedish Research Council Formas (2016-00998). TFK acknowledges support from the Reducing Uncertainties in Biogeochemical Interactions through Synthesis and Computation Scientific Focus Area (RUBISCO SFA), which is sponsored by the Regional and Global Model Analysis (RGMA) Program of the U.S. Department of Energy. YR acknowledges support from National Research Foundation of Korea (NRF-2019R1A2C2084626). This work is a contribution to the LEMONTREE (Land Ecosystem Models based On New Theory, observations and Experiments) project, funded through the generosity of Eric and Wendy Schmidt by recommendation of the Schmidt Futures program (SPH, ICP, HW, HdB, TFK, KTR, YR, NGS, BDS) and to the Imperial College initiative on Grand Challenges in Ecosystems and the Environment (ICP). We thank Belinda Medlyn and Axel Kleidon for helpful comments on an earlier draft of this paper.

Author contributions

SPH, OF, WC, SP, ICP and HW developed the outline of the paper, SPH led the writing process, HW, HdB, NGS, AL, TFK, GM, CM and OF contributed display items, and all authors contributed to the final versions of the paper.

ORCID

Hugo de Boer  <https://orcid.org/0000-0002-6933-344X>
Åke Brännström  <https://orcid.org/0000-0001-9862-816X>
Wolfgang Cramer  <https://orcid.org/0000-0002-9205-5812>
Ulf Dieckmann  <https://orcid.org/0000-0001-7089-0393>
Oskar Franklin  <https://orcid.org/0000-0002-0376-4140>
Sandy P. Harrison  <https://orcid.org/0000-0001-5687-1903>
Jaideep Joshi  <https://orcid.org/0000-0003-1315-6234>

Aliénor Lavergne  <https://orcid.org/0000-0002-4591-1217>
Trevor F. Keenan  <https://orcid.org/0000-0002-3347-0258>
Stefano Manzoni  <https://orcid.org/0000-0002-5960-5712>
Giulia Mengoli  <https://orcid.org/0000-0003-1894-4942>
Catherine Morfopoulos  <https://orcid.org/0000-0002-6121-2483>
Josep Peñuelas  <https://orcid.org/0000-0002-7215-0150>
Stephan Pietsch  <https://orcid.org/0000-0001-6431-2212>
Iain Colin Prentice  <https://orcid.org/0000-0002-1296-6764>
Karin T. Rebel  <https://orcid.org/0000-0002-1722-3935>
Youngryel Ryu  <https://orcid.org/0000-0001-6238-2479>
Nicholas G. Smith  <https://orcid.org/0000-0001-7048-4387>
Benjamin D. Stocker  <https://orcid.org/0000-0003-2697-9096>
Han Wang  <https://orcid.org/0000-0003-2482-1818>
Ian J. Wright  <https://orcid.org/0000-0001-8338-9143>

References

- Ali AA, Xu C, Rogers A, Fisher RA, Wullschlegel SD, Massoud EC, Vrugt JA, Muss JD, McDowell NG, Fisher JB *et al.* 2016. A global scale mechanistic model of photosynthetic capacity (LUNA V1.0). *Geoscientific Model Development* 9: 587–606.
- Anderegg WRL, Wolf A, Arango-Velez A, Choat B, Chmura DJ, Jansen S, Kolb T, Li S, Meinzer FC, Pita P *et al.* 2018. Woody plants optimise stomatal behaviour relative to hydraulic risk. *Ecology Letters* 21: 968–977.
- Arora VK, Katavouta A, Williams RG, Jones CD, Brovkin V, Friedlingstein P, Schwinger J, Bopp L, Boucher O, Cadule P *et al.* 2020. Carbon-concentration and carbon-climate feedbacks in CMIP6 models, and their comparison to CMIP5 models. *Biogeosciences* 17: 4173–4222.
- Atkin OK, Bahar NHA, Bloomfield K, Griffin KL, Heskell MA, Huntingford C, Martinez de la Torre A, Turnbull MH. 2017. Leaf respiration in terrestrial biosphere models. In: Tcherkez G, Ghashghaie J, eds. *Plant respiration: metabolic fluxes and carbon balance*. Cham, Switzerland: Springer International, 107–142.
- Atkin OK, Bloomfield KJ, Reich PB, Tjoelker MG, Asner GP, Bonal D, Bönsch G, Bradford MG, Cernusak LA, Cosio EG *et al.* 2015. Global variability in leaf respiration in relation to climate, plant functional types and leaf traits. *New Phytologist* 206: 614–636.
- Baskaran P, Hyvonen R, Berglund SL, Clemmensen KE, Agren GI, Lindahl BD, Manzoni S. 2017. Modelling the influence of ectomycorrhizal decomposition on plant nutrition and soil carbon sequestration in boreal forest ecosystems. *New Phytologist* 213: 1452–1465.
- Bernacchi CJ, Singaas EL, Pimentel C, Portis AR, Long SP. 2001. Improved temperature response functions for models of Rubisco-limited photosynthesis. *Plant, Cell & Environment* 24: 253–259.
- Bonan GB, Williams M, Fisher RA, Oleson KW. 2014. Modeling stomatal conductance in the earth system: linking leaf water-use efficiency and water transport along the soil-plant-atmosphere continuum. *Geoscientific Model Development* 7: 2193–2222.
- Borer ET, Harpole WS, Adler PB, Lind EM, Orrock JL, Seabloom EW, Smith MD. 2014. Finding generality in ecology: a model for globally distributed experiments. *Methods in Ecology and Evolution* 5: 65–73.
- Brodribb TJ. 2009. Xylem hydraulic physiology: the functional backbone of terrestrial plant productivity. *Plant Science* 177: 245–251.
- Brzostek ER, Fisher JB, Phillips RP. 2014. Modeling the carbon cost of plant nitrogen acquisition: mycorrhizal trade-offs and multipath resistance uptake improve predictions of retranslocation. *Journal of Geophysical Research - Biogeosciences* 119: 1684–1697.
- Buckley TN, John GP, Scoffoni C, Sack L. 2015. How does leaf anatomy influence water transport outside the xylem? *Plant Physiology* 168: 1616–1635.
- Caldararu S, Purves DW, Palmer PI. 2014. Phenology as a strategy for carbon optimality: a global model. *Biogeosciences* 11: 763–778.
- Chen JL, Reynolds JF, Harley PC, Tenhunen JD. 1993. Coordination theory of leaf nitrogen distribution in a canopy. *Oecologia* 93: 63–69.

- Ciais P, Sabine C, Bala G, Bopp L, Brovkin V, Canadell J, Chhabra A, DeFries R, Galloway J, Heimann M *et al.* 2013. Carbon and other biogeochemical cycles. In: Stocker TF, Qin D, Plattner G-K, Tignor M, Allen SK, Boschung J, Nauels A, Xia Y, Bex V, Midgley PM, eds. *Climate change 2013: The physical science basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge, UK: Cambridge University Press, 465–570.
- Clarke PJ, Lawes MJ, Midgley JJ, Lamont BB, Ojeda F, Burrows GE, Enright NJ, Knox KJE. 2013. Resprouting as a key functional trait: how buds, protection and resources drive persistence after fire. *New Phytologist* 197: 19–35.
- Collatz GJ, Ball JT, Grievat C, Berry JA. 1991. Physiological and environmental regulation of stomatal conductance, photosynthesis and transpiration: a model that includes a laminar boundary layer. *Agricultural and Forest Meteorology* 54: 107–136.
- Collier N, Hoffman FM, Lawrence DM, Keppel-Aleks G, Koven CD, Riley WJ, Mu M, Randerson JT. 2018. The International Land Model Benchmarking (ILAMB) system: design, theory, and implementation. *Journal of Advances in Modeling Earth Systems* 10: 2731–2754.
- Cowan I, Farquhar G. 1977. Stomatal function in relation to leaf metabolism and environment: stomatal function in the regulation of gas exchange. *Symposia of the Society for Experimental Biology* 31: 471–505.
- Cox PM, Pearson D, Booth BB, Friedlingstein P, Huntingford C, Jones CD, Luke CM. 2013. Sensitivity of tropical carbon to climate change constrained by carbon dioxide variability. *Nature* 494: 341–344.
- Cui E, Huang K, Arain MA, Fisher JB, Huntzinger DN, Ito A, Luo Y, Jain AK, Mao J, Michalak AM *et al.* 2019. Vegetation functional properties determine uncertainty of simulated ecosystem productivity: a traceability analysis in the East Asian Monsoon Region. *Global Biogeochemical Cycles* 33: 668–689.
- Damour G, Simonneau T, Cochard H, Urban L. 2010. An overview of models of stomatal conductance at the leaf level. *Plant, Cell & Environment* 33: 1419–1438.
- Deans RM, Brodribb TJ, Busch FA, Farquhar GD. 2020. Optimization can provide the fundamental link between leaf photosynthesis, gas exchange and water relations. *Nature Plants* 6: 1116–1125.
- Dewar R, Mauranen A, Mäkelä A, Hölttä T, Medlyn B, Vesala T. 2018. New insights into the covariation of stomatal, mesophyll and hydraulic conductances from optimization models incorporating nonstomatal limitations to photosynthesis. *New Phytologist* 217: 571–585.
- Dieckmann U, Law R. 1996. The dynamical theory of coevolution: a derivation from stochastic ecological processes. *Journal of Mathematical Biology* 34: 579–612.
- Dong N, Prentice IC, Evans BJ, Caddy-Retalic S, Lowe AJ, Wright IJ. 2017. Leaf nitrogen from first principles: field evidence for adaptive variation with climate. *Biogeosciences* 14: 481–495.
- Dong N, Prentice IC, Wright IJ, Evans BJ, Togashi HF, Caddy-Retalic S, McNerney FA, Sparrow B, Leitch E, Lowe AJ. 2020. Components of leaf-trait variation along environmental gradients. *New Phytologist* 228: 82–94.
- Dongsansuk A, Lütz C, Neuner G. 2013. Effects of temperature and irradiance on quantum yield of PSII photochemistry and xanthophyll cycle in a tropical and a temperate species. *Photosynthetica* 51: 13–21.
- Drake JE, Power SA, Duursma RA, Medlyn BE, Aspinwall MJ, Choat B, Creek D, Eamus D, Maier C, Pfautsch S *et al.* 2017. Stomatal and non-stomatal limitations of photosynthesis for four tree species under drought: a comparison of model formulations. *Agricultural and Forest Meteorology* 247: 454–466.
- Dreyer E, Roux XL, Montpied P, Daudet FA, Masson F. 2001. Temperature response of leaf photosynthetic capacity in seedlings from seven temperate tree species. *Tree Physiology* 21: 223–232.
- Dybzinski R, Farrior CE, Pacala SW. 2015. Increased forest carbon storage with increased atmospheric CO₂ despite nitrogen limitation: a game-theoretic allocation model for trees in competition for nitrogen and light. *Global Change Biology* 21: 1182–1196.
- Ehleringer J, Pearcy RW. 1983. Variation in quantum yield for CO₂ uptake among C₃ and C₄ plants. *Plant Physiology* 73: 555–559.
- Eller CB, Rowland L, Oliveira RS, Bittencourt PRL, Barris FV, da Costa ACL, Meir P, Friend AD, Mencuccini M, Sitch S *et al.* 2018. Modelling tropical forest responses to drought and El Niño with a stomatal optimization model based on xylem hydraulics. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* 373: 20170315.
- Eller CB, Rowland L, Mencuccini M, Rosas T, Williams K, Harper A, Medlyn BE, Wagner Y, Klein T, Teodoro GS *et al.* 2020. Stomatal optimization based on xylem hydraulics (SOX) improves land surface model simulation of vegetation responses to climate. *New Phytologist* 226: 1622–1637.
- Falster DS, Brännström Å, Westoby M, Dieckmann U. 2017. Multitrait successional forest dynamics enable diverse competitive coexistence. *Proceedings of the National Academy of Sciences, USA* 114: E2719–E2728.
- Farquhar GD, von Caemmerer S, Berry JA. 1980. A biochemical model of photosynthetic CO₂ assimilation in leaves of C₃ species. *Planta* 149: 78–90.
- Farrior CE, Tilman D, Dybzinski R, Reich PB, Levin SA, Pacala SW. 2013. Resource limitation in a competitive context determines complex plant responses to experimental resource additions. *Ecology* 94: 2505–2517.
- Field C, Mooney HA. 1986. The photosynthesis-nitrogen relationship in wild plants. In: Givnish T, ed. *On the economy of plant form and function*. Cambridge, UK: Cambridge University Press, 25–55.
- Fisher JB, Huntzinger DN, Schwalm CR, Sitch S. 2014. Modeling the terrestrial biosphere. *Annual Review of Environment and Resources* 39: 91–123.
- Fisher RA, Koven CD. 2020. Perspectives on the future of Land Surface Models and the challenges of representing complex terrestrial systems. *Journal of Advances in Modeling Earth Systems* 12: e2018MS001453.
- Forkel M, Andela N, Harrison SP, Lasslop G, van Marle M, Chuvieco E, Dorigo W, Forrest M, Hantson S, Heil A *et al.* 2019. Emergent relationships with respect to burned area in global satellite observations and fire-enabled vegetation models. *Biogeosciences* 16: 57–76.
- Franklin O, Harrison SP, Dewar R, Farrior CE, Brännström A, Dieckmann U, Pietsch S, Falster D, Cramer W, Loreau M *et al.* 2020. Organizing principles for vegetation dynamics. *Nature Plants* 6: 444–453.
- Franklin O, McMurtrie RE, Iversen CM, Crous KY, Finzi AC, Tissue DT, Ellsworth DS, Oren R, Norby RJ. 2009. Forest fine-root production and nitrogen use under elevated CO₂: contrasting responses in evergreen and deciduous trees explained by a common principle. *Global Change Biology* 15: 132–144.
- Franklin O, Näsholm T, Höglberg P, Höglberg MN. 2014. Forests trapped in nitrogen limitation – an ecological market perspective on ectomycorrhizal symbiosis. *New Phytologist* 203: 657–666.
- Franks PJ, Bonan GB, Berry JA, Lombardozzi DL, Holbrook NM, Herold N, Oleson KW. 2018. Comparing optimal and empirical stomatal conductance models for application in Earth system models. *Global Change Biology* 24: 5708–5723.
- Friedlingstein P, Cox P, Betts R, Bopp L, von Bloh W, Brovkin V, Cadule P, Doney S, Eby M, Fung I *et al.* 2006. Climate–carbon cycle feedback analysis: results from the C⁴MIP model intercomparison. *Journal of Climate* 19: 3337–3353.
- Friedlingstein P, O’Sullivan M, Jones MW, Andrew RM, Hauck J, Olsen A, Peters GP, Peters W, Pongratz J, Sitch S *et al.* 2020. Global carbon budget 2020. *Earth System Science Data* 12: 3269–3340.
- Galmés J, Hermida-Carrera C, Laanisto L, Niinemets U. 2016. A compendium of temperature responses of Rubisco kinetic traits: variability among and within photosynthetic groups and impacts on photosynthesis modeling. *Journal of Experimental Botany* 67: 5067–5091.
- Galmés J, Kapralov MV, Copolovici LO, Hermida-Carrera C, Niinemets U. 2015. Temperature responses of the Rubisco maximum carboxylase activity across domains of life: phylogenetic signals, trade-offs, and importance for carbon gain. *Photosynthesis Research* 123: 183–201.
- Gill AL, Finzi AC. 2016. Belowground carbon flux links biogeochemical cycles and resource-use efficiency at the global scale. *Ecological Letters* 12: 1419–1428.
- Gramelsberger G, Lenhard J, Parker WS. 2020. Philosophical perspectives on Earth system modeling: truth, adequacy, and understanding. *Journal of Advances in Modeling Earth Systems* 12: e2019MS001720.
- Graven H, Keeling RF, Piper SC, Patra PK, Stephens BB, Wofsy SC, Welp LR, Sweeney C, Tans PP, Kelley JJ *et al.* 2013. Enhanced seasonal exchange of CO₂ by northern ecosystems since 1960. *Science* 341: 1085–1089.
- Guenther A, Karl T, Harley P, Wiedinmyer C, Palmer PI, Geron C. 2006. Estimates of global terrestrial isoprene emissions using MEGAN (Model of

- Emissions of Gases and Aerosols from Nature. *Atmospheric Chemistry and Physics* 6: 3181–3210.
- Hanson S, Kelley DI, Arneth A, Harrison SP, Archibald S, Bachelet D, Forrest M, Kloster S, Lasslop G, Li F *et al.* 2020. Quantitative assessment of fire and vegetation properties in historical simulations with fire-enabled vegetation models from the FireMIP intercomparison project. *Geoscientific Model Development* 13: 3299–3318.
- Harper AB, Cox PM, Friedlingstein P, Wiltshire AJ, Jones CD, Sitch S, Mercado LM, Groenendijk M, Robertson E, Kattge J *et al.* 2016. Improved representation of plant functional types and physiology in the Joint UK Land Environment Simulator (JULES v4.2) using plant trait information. *Geoscientific Model Development* 9: 2415–2440.
- Harrison SP, Morfopoulos C, Dani KG, Prentice IC, Arneth A, Atwell BJ, Barkley MP, Leishman MR, Loreto F, Medlyn BE *et al.* 2013. Volatile isoprenoid emissions from plastid to planet. *New Phytologist* 197: 49–57.
- Harrison SP, Sanchez Goñi MF. 2010. Global patterns of vegetation response to millennial-scale variability during the last glacial: a synthesis. *Quaternary Science Reviews* 29: 2957–2980.
- Heck V, Hoff H, Wirsén S, Meyer C, Kreft H. 2018. Land use options for staying within the Planetary Boundaries – Synergies and trade-offs between global and local sustainability goals. *Global Environmental Change* 49: 73–84.
- Held IM. 2005. The gap between simulation and understanding in climate modeling. *Bulletin of the American Meteorological Society* 86: 1609–1614.
- Heskel MA, O'Sullivan OS, Reich PB, Tjoelker MG, Weerasinghe LK, Penillard A, Egerton JJG, Creek D, Bloomfield KJ, Xiang J *et al.* 2016. Convergence in the temperature response of leaf respiration across biomes and plant functional types. *Proceedings of the National Academy of Sciences, USA* 113: 3832–3837.
- Hilton TW, Loik ME, Campbell JE. 2019. Simulating International Drought Experiment field observations using the community land model. *Agricultural and Forest Meteorology* 266–267: 173–183.
- Hoepfner SS, Dukes JS. 2012. Interactive responses of old-field plant growth and composition to warming and precipitation. *Global Change Biology* 18: 1754–1768.
- Hovenden MJ, Leuzinger S, Newton PCD, Fletcher A, Fatichi S, Lüscher A, Reich PB, Andresen LC, Beier C, Blumenthal DM *et al.* 2019. Globally consistent influences of seasonal precipitation limit grassland biomass response to elevated CO₂. *Nature Plants* 5: 167–173.
- Huntingford C, Atkin OK, Martinez-de la Torre A, Mercado LM, Mary A H, Harper AB, Bloomfield KJ, O'Sullivan OS, Reich PB, Wythers KR *et al.* 2017. Implications of improved representations of plant respiration in a changing climate. *Nature Communications* 8: 1602.
- Huntzinger DN, Michalak AM, Schwalm C, Ciais P, King AW, Fang FY, Schaefer K, Wei Y, Cook RB, Fisher JB *et al.* 2017. Uncertainty in the response of terrestrial carbon sink to environmental drivers undermines carbon-climate feedback predictions. *Scientific Reports* 7: 4765.
- Inoue Y, Ichie T, Kenzo T, Yoneyama A, Kumagai T, Nakashizuka T. 2017. Effects of rainfall exclusion on leaf gas exchange traits and osmotic adjustment in mature canopy trees of *Dryobalanops aromatica* (Dipterocarpaceae) in a Malaysian tropical rain forest. *Tree Physiology* 37: 1301–1311.
- Jiang C, Ryu Y, Wang H, Keenan TF. 2020. An optimality-based model explains seasonal variation in C₃ plant photosynthetic capacity. *Global Change Biology* 26: 6493–6510.
- Joshi J, Stocker BD, Hofhansl F, Zhou S, Dieckmann U, Prentice IC. 2020. Towards a unified theory of plant photosynthesis and hydraulics. *bioRxiv*. doi: 10.1101/2020.12.17.423132.
- Kattge J *et al.* 2020. Twelve years of TRY – towards a third generation of plant trait data assimilation and sharing. *Global Change Biology* 26: 119–188.
- De Kauwe MG, Kala J, Lin YS, Pitman AJ, Medlyn BE, Duursma RA, Abramowitz G, Wang YP, Miralles DG. 2015. A test of an optimal stomatal conductance scheme within the CABLE land surface model. *Geoscientific Model Development* 8: 431–452.
- Keenan TF, Prentice IC, Canadell JG, Williams CA, Wang H, Raupach M, Collatz GJ. 2016. Recent pause in the growth rate of atmospheric CO₂ due to enhanced terrestrial carbon uptake. *Nature Communications* 7: 13428.
- Kikuzawa K. 1991. A cost-benefit analysis of leaf habit and leaf longevity of trees and their geographical pattern. *The American Naturalist* 138: 1250–1263.
- Klein Goldewijk K, Beusen A, Doelman J, Stehfest E. 2017. Anthropogenic land-use estimates for the Holocene; HYDE 3.2. *Earth System Science Data* 9: 927–953.
- Knapp AK, Avolio ML, Beier C, Carroll CJW, Collins SL, Dukes JS, Fraser LH, Griffin-Nolan RJ, Hoover DL, Jentsch A *et al.* 2017. Pushing precipitation to the extremes in distributed experiments: recommendations for simulating wet and dry years. *Global Change Biology* 23: 1774–1782.
- Körner C, Diemer M. 1987. In situ photosynthetic responses to light, temperature and carbon dioxide in herbaceous plants from low and high altitude. *Functional Ecology* 1: 179–194.
- Kumarathunge DP, Medlyn BE, Drake JE, Tjoelker MG, Aspinwall MJ, Battaglia M, Cano FJ, Carter KR, Cavaleri MA, Cernusak LA *et al.* 2019. Acclimation and adaptation components of the temperature dependence of plant photosynthesis at the global scale. *New Phytologist* 222: 768–784.
- Kvakić M, Tzagkarakis G, Pellerin S, Ciais P, Goll D, Mollier A, Ringeval B. 2020. Carbon and phosphorus allocation in annual plants: an optimal functioning approach. *Frontiers in Plant Science* 11: 149.
- Lantz AT, Allman J, Weraduwa SM, Sharkey TD. 2019. Isoprene: new insights into the control of emission and mediation of stress tolerance by gene expression. *Plant, Cell & Environment* 42: 2808–2826.
- Laverne A, Sandoval D, Hare VJ, Graven H, Prentice IC. 2020a. Impacts of soil water stress on the acclimated stomatal limitation of photosynthesis: insights from stable carbon isotope data. *Global Change Biology* 26: 7158–7172.
- Laverne A, Voelker S, Csank A, Graven H, de Boer HJ, Daux V, Robertson I, Dorado-Liñán I, Martínez-Sancho E, Battipaglia G *et al.* 2020b. Historical changes in the stomatal limitation of photosynthesis: empirical support for an optimality principle. *New Phytologist* 225: 2484–2497.
- Lawrence DM, Fisher RA, Koven CD, Oleson KW, Swenson SC, Bonan G, Collier N, Ghimire B, Kampenhout L, Kennedy D *et al.* 2019. The Community Land Model version 5: description of new features, benchmarking, and impact of forcing uncertainty. *Journal of Advances in Modeling Earth Systems* 11: 4245–4287.
- Lin Y-S, Medlyn BE, Duursma RA, Prentice IC, Wang H, Baig S, Eamus D, de Dios VR, Mitchell P, Ellsworth DS *et al.* 2015. Optimal stomatal behaviour around the world. *Nature Climate Change* 5: 459–464.
- Lindh M, Zhang L, Falster D, Franklin O, Brännström Å. 2014. Plant diversity and drought: the role of deep roots. *Ecological Modelling* 290: 85–93.
- Locke AM, Sack L, Bernacchi CJ, Ort DR. 2013. Soybean leaf hydraulic conductance does not acclimate to growth at elevated [CO₂] or temperature in growth chambers or in the field. *Annals of Botany* 112: 911–918.
- Lusk CH, Clearwater MJ, Laughlin DC, Harrison SP, Prentice IC, Nordenstahl M, Smith B. 2018. Frost and leaf-size gradients in forests: global patterns and experimental evidence. *New Phytologist* 29: 565–573.
- Maire V, Martre P, Kattge J, Gastal F, Esser G, Fontaine S, Soussana J-F. 2012. The coordination of leaf photosynthesis links C and N fluxes in C₃ plant species. *PLoS One* 7: e38345.
- Manzoni S, Vico G, Palmroth S, Porporato A, Katul G. 2013. Optimization of stomatal conductance for maximum carbon gain under dynamic soil moisture. *Advances in Water Resources* 62: 90–105.
- Manzoni S, Vico G, Thompson S, Beyer F, Weih M. 2015. Contrasting leaf phenological strategies optimize carbon gain under droughts of different duration. *Advances in Water Resources* 84: 37–51.
- Marchin RM, Broadhead AA, Bostic LE, Dunn RR, Hoffmann WA. 2016. Stomatal acclimation to vapour pressure deficit doubles transpiration of small tree seedlings with warming. *Plant, Cell & Environment* 39: 2221–2234.
- Medlyn BE, Duursma RA, De Kauwe MG, Prentice IC. 2013. The optimal stomatal response to atmospheric CO₂ concentration: alternative solutions, alternative interpretations. *Agriculture and Forest Meteorology* 182–183: 200–203.
- Medlyn BE, Duursma RA, Eamus D, Ellsworth DS, Prentice IC, Barton CVM, Crous KY, De Angelis P, Freeman M, Wingate L. 2011. Reconciling the optimal and empirical approaches to modelling stomatal conductance. *Global Change Biology* 17: 2134–2144.
- Medlyn BE, Zaehle S, De Kauwe MG, Walker AP, Dietze MC, Hanson PJ, Hickler T, Jain AK, Luo Y, Panton W *et al.* 2015. Using ecosystem experiments to improve vegetation models. *Nature Climate Change* 5: 528–534.
- Meng T-T, Wang H, Harrison SP, Prentice IC, Ni J, Wang G. 2015. Responses of leaf traits to climatic gradients: adaptive variation versus compositional shifts. *Biogeosciences* 12: 5339–5352.

- Mengoli G, Agusti-Panareda A, Boussetta S, Harrison SP, Trotta C, Prentice IC. 2021. Ecosystem photosynthesis in land-surface models: a first-principles approach. *bioRxiv*. doi: 10.1101/2021.05.07.442894.
- Metz JAJ, Mylius SD, Diekmann O. 2008. When does evolution optimize? *Evolutionary Ecology Research* 10: 629–654.
- Metz JAJ, Nisbet RM, Geritz SAH. 1992. How should we define 'fitness' for general ecological scenarios? *Trends in Ecology and Evolution* 7: 198–202.
- Morfopoulos C. 2014. *A unifying model for isoprene emission by plants*. PhD dissertation. Imperial College of London, London, UK. [WWW document] URL <https://spiral.imperial.ac.uk/handle/10044/1/25115> [accessed 1 May 2021].
- Morfopoulos C, Prentice IC, Keenan TF, Friedlingstein P, Medlyn BE, Peñuelas J, Possell M. 2013. A unifying conceptual model for the environmental responses of isoprene emission by plants. *Annals of Botany* 112: 1223–1238.
- Morfopoulos C, Sperlich D, Peñuelas J, Cubells JF, Llusà J, Medlyn BE, Possell M, Prentice IC. 2014. A model of plant isoprene emission based on available reducing power captures responses to atmospheric CO₂. *New Phytologist* 203: 125–139.
- Niinemets Ü. 2010. Mild versus severe stress and BVOCs: thresholds, priming and consequences. *Trends in Plant Science* 15: 145–153.
- Oliver RJ, Mercado LM, Sitch S, Simpson D, Medlyn BE, Lin Y-S, Folberth GA. 2018. Large but decreasing effect of ozone on the European carbon sink. *Biogeosciences* 15: 4245–4269.
- Ostberg S, Boysen LR, Schaphoff S, Lucht W, Gerten D. 2018. The biosphere under potential Paris outcomes. *Earth's Future* 6: 23–39.
- Pacifico F, Harrison SP, Jones CD, Arneth A, Sitch S, Weedon G, Barkley MP, Palmer PI, Serça D, Potosnak M *et al.* 2011. Evaluation of a photosynthesis-based biogenic isoprene emission scheme in JULES and simulation of isoprene emissions under present-day climate conditions. *Atmospheric Chemistry and Physics* 11: 4371–4389.
- Paillasa J, Wright IJ, Prentice IC, Pepin S, Smith NG, Ethier G, Westerband AC, Lamarque LJ, Wang H, Cornwell WK *et al.* 2020. When and where soil is important to modify the carbon and water economy of leaves. *New Phytologist* 228: 121–135.
- Pastorello G, Trotta C, Canfora E, Chu H, Christianson D, Cheah YW, Poindexter C, Chen J, Elbashandy A, Humphrey M *et al.* 2020. The FLUXNET2015 dataset and the ONEFlux processing pipeline for eddy covariance data. *Scientific Data* 7: 225.
- Pausas JG. 2019. Generalized fire response strategies in plants and animals. *Oikos* 128: 147–153.
- Pausas JG, Pratt RB, Keeley JE, Jacobsen AL, Ramirez AR, Vilagrosa A, Paula S, Kaneakua-Pia IN, Davis SD. 2016. Towards understanding resprouting at the global scale. *New Phytologist* 209: 945–954.
- Peng Y, Bloomfield KJ, Prentice IC. 2020. A theory of plant function helps to explain leaf-trait and productivity responses to elevation. *New Phytologist* 226: 1274–1284.
- Peters W, van der Velde IR, van Schaik E, Miller JB, Ciaia P, Duarte HF, van der Laan-Luijckx IT, van der Molen MK, Scholze M, Schaefer K *et al.* 2018. Increased water-use efficiency and reduced CO₂ uptake by plants during droughts at a continental scale. *Nature Geosciences* 11: 744–748.
- Pettinari ML, Chuvieco E. 2016. Generation of a global fuel data set using the Fuel Characteristic Classification System. *Biogeosciences* 13: 2061–2076.
- Pongratz J, Reick C, Raddatz T, Claussen M. 2008. A re-construction of global agricultural areas and land cover for the last millennium. *Global Biogeochemical Cycles* 22: GB3018.
- Poorter H, Niklas KJ, Reich PB, Oleksyn J, Poot P, Mommer L. 2012. Biomass allocation to leaves, stems and roots: meta-analyses of interspecific variation and environmental control. *New Phytologist* 193: 30–50.
- Prentice IC, Bondeau A, Cramer W, Harrison SP, Hickler T, Lucht W, Sitch S, Smith B, Sykes MT. 2007. Dynamic global vegetation modelling: quantifying terrestrial ecosystem responses to large-scale environmental change. In: Canadell J, Pitelka L, Pataki D, eds. *Terrestrial ecosystems in a changing world*. Berlin, Germany: Springer-Verlag, 175–192.
- Prentice IC, Cowling SA. 2013. Dynamic global vegetation models. In: Levin SA, ed. *Encyclopedia of biodiversity*, 2nd edn, vol. 2. Cambridge, MA, USA: Academic Press, 607–689.
- Prentice IC, Dong N, Gleason SM, Maire V, Wright IJ. 2014. Balancing the costs of carbon gain and water transport: testing a new theoretical framework for plant functional ecology. *Ecology Letters* 17: 82–91.
- Prentice IC, Liang X, Medlyn BE, Wang YP. 2015. Reliable, robust and realistic: the three R's of next-generation land-surface modelling. *Atmospheric Chemistry & Physics* 15: 5987–6005.
- Qiao S, Wang H, Prentice IC, Harrison SP. 2020. Extending a generic primary production model to predict wheat yield. *Agriculture and Forest Meteorology* 287: 107932.
- Quebbeman JA, Ramirez JA. 2016. Optimal allocation of leaf-level nitrogen: implications for covariation of V_{cmax} and J_{cmax} and photosynthetic downregulation. *Journal of Geophysical Research Biogeosciences* 121: 2464–2475.
- Quillet A, Peng C, Garneau M. 2010. Toward dynamic global vegetation models for simulating vegetation–climate interactions and feedbacks: recent developments, limitations, and future challenges. *Environmental Reviews* 18: 333–353.
- Rastetter EB, Kwiatkowski BL. 2020. An approach to modeling resource optimization for substitutable and interdependent resources. *Ecological Modelling* 425: 109033.
- Rastetter EB, Shaver GR. 1992. A model of multiple-element limitation for acclimating vegetation. *Ecology* 73: 1157–1174.
- Reich PB, Luo Y, Bradford JB, Poorter H, Perry CH, Oleksyn J. 2014. Temperature drives forest biomass distribution. *Proceedings of the National Academy of Sciences, USA* 111: 13721–13726.
- Reichstein M, Bahn M, Mahecha MD, Kattge J, Baldocchi DD. 2014. Linking plant and ecosystem functional biogeography. *Proceedings of the National Academy of Sciences, USA* 111: 13697–13702.
- Rogers A, Medlyn BE, Dukes JS, Bonan G, Caemmerer S, Dietze MC, Kattge J, Leakey ADB, Mercado LM, Niinemets Ü *et al.* 2017. A roadmap for improving the representation of photosynthesis in Earth system models. *New Phytologist* 213: 22–42.
- Sabot MEB, de Kauwe MG, Pitman AJ, Medlyn BE, Verhoef A, Ukkola AM, Abramowitz G. 2020. Plant profit maximization improves predictions of European forest responses to drought. *New Phytologist* 226: 1638–1655.
- Sakschewski B, von Bloh W, Boit A, Rammig A, Kattge J, Poorter L, Peñuelas J, Thonicke K. 2015. Leaf and stem economics spectra drive diversity of functional plant traits in a dynamic global vegetation model. *Global Change Biology* 21: 2711–2725.
- Scafaro AP, Xiang S, Long BM, Bahar NHA, Weerasinghe LK, Creek D, Evans JR, Reich PN, Atkin OK. 2017. Strong thermal acclimation of photosynthesis in tropical and temperate wet-forest tree species: the importance of altered Rubisco content. *Global Change Biology* 23: 2783–2800.
- Shi M, Fisher JB, Brzostek ER, Phillips RP. 2016. Carbon cost of plant nitrogen acquisition: global carbon cycle impact from an improved plant nitrogen cycle in the Community Land Model. *Global Change Biology* 22: 1299–1314.
- Siefert A, Violle C, Chalmandrier L, Albert CH, Taudiere A, Fajardo A, Aarssen LW, Baraloto C, Carlucci MB, Cianciaruso MV *et al.* 2015. A global meta-analysis of the relative extent of intraspecific trait variation in plant communities. *Ecology Letters* 18: 1406–1419.
- Singhania EL, Ort DR, DeLucia EH. 2001. Variation in measured values of photosynthetic quantum yield in ecophysiological studies. *Oecologia* 128: 15–23.
- Smith NG, Dukes JS. 2017. LCE: leaf carbon exchange data set for tropical, temperate, and boreal species of North and Central America. *Ecology* 98: 2978.
- Smith NG, Dukes JS. 2018. Drivers of leaf carbon exchange capacity across biomes at the continental scale. *Ecology* 99: 1610–1620.
- Smith NG, Keenan TF, Prentice IC, Wang H, Wright IJ, Niinemets Ü, Crous Y, Domingues TF, Guerrieri R, Ishida FY *et al.* 2019. Global photosynthetic capacity is optimized to the environment. *Ecology Letters* 22: 506–517.
- Smith NG, Keenan TF. 2020. Mechanisms underlying leaf photosynthetic acclimation to warming and elevated CO₂ as inferred from least-cost optimality theory. *Global Change Biology* 26: 5202–5216.
- Sperry JS, Venturas MD, Anderegg WRL, Mencuccini M, Mackay DS, Wang Y, Love DM. 2017. Predicting stomatal responses to the environment from the optimization of photosynthetic gain and hydraulic cost. *Plant, Cell & Environment* 40: 816–830.
- Stocker BD, Wang H, Smith NG, Harrison SP, Keenan T, Sandoval D, Davis T, Prentice IC. 2020. P-model v1.0: an optimality-based light use efficiency model

- for terrestrial gross primary production. *Geoscientific Model Development* 13: 1545–1581.
- Sulman BN, Brzostek ER, Medici C, Shevliakova E, Menge DNL, Phillips RP. 2017. Feedbacks between plant N demand and rhizosphere priming depend on type of mycorrhizal association. *Ecology Letters* 20: 1043–1053.
- Sun Z, Hüve K, Vislap V, Niinemets Ü. 2013. Elevated [CO₂] magnifies isoprene emissions under heat and improves thermal resistance in hybrid aspen. *Journal of Experimental Botany* 64: 5509–5523.
- Taylor TC, McMahon SM, Smith MN, Boyle B, Violle C, van Haren J, Simova I, Meir P, Ferreira LV, de Camargo PB *et al.* 2018. Isoprene emission structures tropical tree biogeography and community assembly responses to climate. *New Phytologist* 220: 435–446.
- Terrer C, Vicca S, Stocker BD, Hungate BA, Phillips RP, Reich PB, Finzi AC, Prentice IC. 2018. Ecosystem responses to elevated CO₂ governed by plant–soil interactions and the cost of nitrogen acquisition. *New Phytologist* 217: 507–522.
- Thomas RB, Prentice IC, Graven H, Ciais P, Fisher JB, Huang M, Huntzinger DN, Ito A, Jacobson A, Jain A *et al.* 2016. Increased light-use efficiency in northern terrestrial ecosystems indicated by CO₂ and greening observations. *Geophysical Research Letters* 43: 11339–11349.
- Thomas RQ, Williams M. 2014. A model using marginal efficiency of investment to analyse carbon and nitrogen interactions in terrestrial ecosystems (ACONITE Version 1). *Geoscientific Model Development* 7: 2015–2037.
- Togashi HF, Atkin OK, Bloomfield KJ, Bradford M, Cao K, Dong N, Evans BJ, Fan Z, Harrison SP, Hua Z *et al.* 2018a. Functional trait variation related to gap dynamics in tropical moist forests: a vegetation modelling perspective. *Perspectives in Plant Ecology, Evolution and Systematics* 35: 52–64.
- Togashi HF, Prentice IC, Atkin OK, Macfarlane C, Prober SM, Bloomfield KJ, Evans BJ. 2018b. Thermal acclimation of leaf photosynthetic traits to temperature in an evergreen woodland, consistent with the coordination hypothesis. *Biogeosciences* 15: 3461–3474.
- Tomasella M, Beikircher B, Häberle K-H, Hesse B, Kallenbach C, Matyssek R, Mayr S. 2018. Acclimation of branch and leaf hydraulics in adult *Fagus sylvatica* and *Picea abies* in a forest through-fall exclusion experiment. *Tree Physiology* 38: 198–211.
- Tramontana G, Jung M, Schwalm CR, Ichii K, Camps-Valls G, Raduly B, Reichstein M, Arain MA, Cescatti A, Kiely G *et al.* 2016. Predicting carbon dioxide and energy fluxes across global FLUXNET sites with regression algorithms. *Biogeosciences* 13: 4291–4313.
- Trugman AT, Anderegg LDL, Sperry JS, Wang Y, Venturas M, Anderegg WRL. 2019. Leveraging plant hydraulics to yield predictive and dynamic plant leaf allocation in vegetation models with climate change. *Global Change Biology* 25: 4008–4021.
- Ukkola AM, Prentice IC. 2013. A worldwide analysis of trends in water-balance evapotranspiration. *Hydrology & Earth System Science* 17: 4177–4187.
- Valentine HT, Mäkelä A. 2012. Modeling forest stand dynamics from optimal balances of carbon and nitrogen. *New Phytologist* 194: 961–971.
- Van Bodegom PM, Douma JC, Witte JPM, Ordoñez JC, Bartholomeus RP, Aerts R. 2012. Going beyond limitations of plant functional types when predicting global ecosystem-atmosphere fluxes: exploring the merits of traits-based approaches. *Global Ecology & Biogeography* 21: 625–636.
- VEMAP. 1995. Vegetation/ecosystem modeling and analysis project: comparing biogeography and biogeochemistry models in a continental-scale study of terrestrial ecosystem responses to climate change and CO₂ doubling. *Global Biogeochemical Cycles* 9: 407–437.
- Venturas MD, Sperry JS, Love DM, Frehner EH, Allred MG, Wang Y, Anderegg WRL. 2018. A stomatal control model based on optimization of carbon gain versus hydraulic risk predicts aspen sapling responses to drought. *New Phytologist* 220: 836–850.
- Verheijen LM, Aerts R, Brovkin V, Cavender-Bares J, Cornelissen JHC, Kattge J, van Bodegom PM. 2015. Inclusion of ecologically based trait variation in plant functional types reduces the projected land carbon sink in an Earth system model. *Global Change Biology* 21: 3074–3086.
- Vico G, Way DA, Hurry V, Manzoni S. 2019. Can leaf net photosynthesis acclimate to rising and more variable temperatures? *Plant, Cell, & Environment* 42: 1913–1928.
- Wang H, Atkin OK, Keenan TF, Smith NG, Wright IJ, Bloomfield KJ, Kattge J, Reich PB, Prentice IC. 2020. Acclimation of leaf respiration consistent with optimal photosynthetic capacity. *Global Change Biology* 26: 2573–2583.
- Wang H, Prentice IC, Cornwell WM, Keenan TF, Davis TW, Wright IJ, Evans BJ, Peng C. 2017. Towards a universal model for carbon dioxide uptake by plants. *Nature Plants* 3: 734–741.
- Wang H, Prentice IC, Davis TW. 2014. Biophysical constraints on gross primary production by the terrestrial biosphere. *Biogeosciences* 11: 5987–6001.
- Wang H, Prentice IC, Wright IJ, Qiao S, Xu X, Kikuzawa K, Stenseth NC. 2021. Leaf economics explained by optimality principles. *bioRxiv*: doi: 10.1101/2021.02.07.430028.
- Wang Y, Sperry JS, Anderegg WRL, Venturas MD, Trugman AT. 2020. A theoretical and empirical assessment of stomatal optimization modeling. *New Phytologist* 227: 311–325.
- Weng E, Dybzinski R, Farrior CE, Pacala SW. 2019. Competition alters predicted forest carbon cycle responses to nitrogen availability and elevated CO₂: simulations using an explicitly competitive, game-theoretic vegetation demographic model. *Biogeosciences* 16: 4577–4599.
- Wieder WR, Cleveland CC, Smith WK, Todd-Brown K. 2015. Future productivity and carbon storage limited by terrestrial nutrient availability. *Nature Geosciences* 8: 441–444.
- Wieder WR, Lawrence DM, Fisher RA, Bonan GB, Cheng SJ, Goodale CL, Grandy AS, Koven CD, Lombardozzi DL, Oleson KW *et al.* 2019. Beyond static benchmarking: using experimental manipulations to evaluate land model assumptions. *Global Biogeochemical Cycles* 33: 1289–1309.
- Wolf A, Anderegg WRL, Pacala SW. 2016. Optimal stomatal behavior with competition for water and risk of hydraulic impairment. *Proceedings of the National Academy of Sciences, USA* 113: E7222–E7230.
- Wright IJ, Reich PB, Westoby M. 2003. Least-cost input mixtures of water and nitrogen for photosynthesis. *The American Naturalist* 161: 98–111.
- Wright IJ, Reich PB, Westoby M, Ackerly DD, Baruch Z, Bongers F, Cavender-Bares J, Chapin T, Cornelissen JHC, Diemer M *et al.* 2004. The worldwide leaf economics spectrum. *Nature* 428: 821–827.
- Xia J, Yuan W, Wang YP, Zhang Q. 2017. Adaptive carbon allocation by plants enhances the terrestrial carbon sink. *Scientific Reports* 7: 3341.
- Xu H, Wang H, Prentice IC, Harrison SP, Wang G, Sun X. 2021. Predictability of leaf traits with climate and elevation: a case study in Gongga Mountain, China. *Tree Physiology*. doi: 10.1093/treephys/tpab003.
- Xu X, Medvigy D, Wright JS, Kitajima K, Wu J, Albert LP, Martins GA, Saleska SR, Pacala SW. 2017. Variations of leaf longevity in tropical moist forests predicted by a trait driven carbon optimality model. *Ecology Letters* 20: 1097–1106.
- Yang J, Cao M, Swenson NG. 2018. Why functional traits do not predict tree demographic rates. *Trends in Ecology & Evolution* 33: 326–336.

Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Methods S1 Description of experimental treatment.

Table S1 Responses of maximum carboxylation capacity to experimental treatments.

Please note: Wiley Blackwell are not responsible for the content or functionality of any Supporting Information supplied by the authors. Any queries (other than missing material) should be directed to the *New Phytologist* Central Office.